

# Native Plant Seed Sowing Guide

Western Australian native species – Joondalup, WA

Version 1.2 · September 2026 · 41 species

## Species in this edition (alphabetical)

Full index on page 2

|                                    |    |                                   |    |                                  |    |
|------------------------------------|----|-----------------------------------|----|----------------------------------|----|
| ● <i>Allocasuarina fraseriana</i>  | 7  | ● <i>Daviesia nudiflora</i>       | 35 | ● <i>Olearia axillaris</i>       | 63 |
| ● <i>Allocasuarina humilis</i>     | 9  | ● <i>Daviesia triflora</i>        | 37 | ● <i>Petrophile linearis</i>     | 65 |
| ● <i>Allocasuarina lehmanniana</i> | 11 | ● <i>Dianella revoluta</i>        | 39 | ● <i>Petrophile macrostachya</i> | 67 |
| ● <i>Anigozanthos manglesii</i>    | 13 | ● <i>Eucalyptus gomphocephala</i> | 41 | ● <i>Podotheca gnaphalioides</i> | 69 |
| ● <i>Atriplex isatidea</i>         | 15 | ● <i>Eucalyptus marginata</i>     | 43 | ● <i>Ptilotus drummondii</i>     | 71 |
| ● <i>Austrostipa flavescens</i>    | 17 | ● <i>Gompholobium tomentosum</i>  | 45 | ● <i>Ptilotus manglesii</i>      | 73 |
| ● <i>Banksia attenuata</i>         | 19 | ● <i>Grevillea vestita</i>        | 47 | ● <i>Ptilotus polystachyus</i>   | 75 |
| ● <i>Banksia menziesii</i>         | 21 | ● <i>Hovea pungens</i>            | 49 | ● <i>Rhagodia baccata</i>        | 77 |
| ● <i>Banksia sessilis</i>          | 23 | ● <i>Kennedia prostrata</i>       | 51 | ● <i>Spinifex hirsutus</i>       | 79 |
| ● <i>Bossiaea eriocarpa</i>        | 25 | ● <i>Leucophyta brownii</i>       | 53 | ● <i>Spinifex longifolius</i>    | 81 |
| ● <i>Calothamnus quadrifidus</i>   | 27 | ● <i>Macrozamia fraseri</i>       | 55 | ● <i>Spyridium globulosum</i>    | 83 |
| ● <i>Carpobrotus virescens</i>     | 29 | ● <i>Melaleuca huegelii</i>       | 57 | ● <i>Threlkeldia diffusa</i>     | 85 |
| ● <i>Corymbia calophylla</i>       | 31 | ● <i>Melaleuca systema</i>        | 59 | ● <i>Waitzia suaveolens</i>      | 87 |
| ● <i>Daviesia divaricata</i>       | 33 | ● <i>Myoporum insulare</i>        | 61 |                                  |    |

● Easy ● Easy-Mod. ● Moderate ● Mod.-Difficult ● Difficult

Germination figures are nursery test averages – your own seed lots will vary. Record results to refine this guide.

Compiled from published sources (see References). Windows and ratings are starting points for local practice.

# How to use this guide

Each species has its own page set, listed alphabetically by scientific name. Every page follows the same layout: quick facts, a sowing calendar, a plant snapshot, seed collection and storage, the sowing and growing-on method, common problems, and a blank record table so the nursery can log germination and refine the advice over time.

Sowing windows are written for the Joondalup / Perth coastal climate. Most south-west WA seed germinates best at cool to mild temperatures (roughly 15–25°C), so autumn is the main sowing season. Germination figures come from routine tests at Alcoa's Marrinup Nursery and are a guide to what a good seed lot can do – not a guarantee.

## Species index – at a glance

| Species                                                      | Page | Difficulty    | Pre-treatment                    | Best sowing          | Depth      | Germ. %* | Seed / 100 plants* |
|--------------------------------------------------------------|------|---------------|----------------------------------|----------------------|------------|----------|--------------------|
| <i>Allocasuarina fraseriana</i><br>Sheoak / Western Sheoak   | 7    | Easy          | None                             | Mar – May            | 2–3 mm     | 37%      | 1.3 g              |
| <i>Allocasuarina humilis</i><br>Dwarf Sheoak                 | 9    | Easy-Moderate | None                             | Mar – May            | 2–3 mm     | 25%      | 1.3 g              |
| <i>Allocasuarina lehmanniana</i><br>Dune Sheoak              | 11   | Easy-Moderate | None                             | Mar – May            | 2–3 mm     | 19%      | 0.5 g              |
| <i>Anigozanthos manglesii</i><br>Red and Green Kangaroo Paw  | 13   | Moderate      | Dry heat 100°C 3 h, or aged seed | Mar – May            | 1–2 mm     | 70%      | 0.5 g              |
| <i>Atriplex isatidea</i><br>Coast Saltbush                   | 15   | Easy-Moderate | Soak 4 days; smoke water         | Mar – May            | 1–2 mm     | 22%      | 6.7 g              |
| <i>Austrostipa flavescens</i><br>Coast Speargrass            | 17   | Easy-Moderate | Smoke water; aged seed           | Mar – May            | 0–2 mm     | –        | –                  |
| <i>Banksia attenuata</i><br>Slender Banksia / Candle Banksia | 19   | Easy-Moderate | None (extract from cones)        | Mar – May            | 3–5 mm     | 60%      | 6.7 g              |
| <i>Banksia menziesii</i><br>Firewood Banksia                 | 21   | Easy-Moderate | None (extract from cones)        | Mar – May            | 3–5 mm     | 56%      | 3.7 g              |
| <i>Banksia sessilis</i><br>Parrot Bush                       | 23   | Easy-Moderate | None (smoke water optional)      | Mar – May            | 3–5 mm     | 42%      | 3.0 g              |
| <i>Bossiaea eriocarpa</i><br>Common Brown Pea                | 25   | Easy-Moderate | Boiling water 30 s               | Mar – May            | 2–3 mm     | 44%      | 0.4 g              |
| <i>Calothamnus quadrifidus</i><br>One-sided Bottlebrush      | 27   | Easy          | None                             | Mar – Apr, Sep – Oct | Surface    | 26%      | 0.2 g              |
| <i>Carpobrotus virescens</i><br>Coastal Pigface              | 29   | Easy-Moderate | Ferment fruit; smoke water       | Mar – May            | Surface    | 21%      | 0.2 g              |
| <i>Corymbia calophylla</i><br>Marri                          | 31   | Easy          | None                             | Mar – Apr, Sep – Oct | About 5 mm | 60%      | 5.3 g              |
| <i>Daviesia divaricata</i><br>Marno                          | 33   | Moderate      | Boiling water 30 s               | Mar – May            | 3–5 mm     | 28%      | 9.2 g              |
| <i>Daviesia nudiflora</i>                                    | 35   | Moderate      | Boiling water 30 s               | Mar – May            | 3–5 mm     | 22%      | 1.8 g              |
| <i>Daviesia triflora</i>                                     | 37   | Moderate      | Boiling water 30 s               | Mar – May            | 3–5 mm     | 26%      | 9.2 g              |
| <i>Dianella revoluta</i><br>Blueberry Lily / Flax Lily       | 39   | Difficult     | GA 24 h + smoke water 24 h       | Mar – Apr            | 3–5 mm     | <1%†     | –                  |
| <i>Eucalyptus gomphocephala</i><br>Tuart                     | 41   | Easy          | None                             | Mar – Apr, Sep – Oct | 1–2 mm     | 42%      | 0.4 g              |
| <i>Eucalyptus marginata</i><br>Jarrah                        | 43   | Easy          | None                             | Mar – Apr, Sep – Oct | 2–3 mm     | 31%      | 3.0 g              |

| Species                                                              | Page | Difficulty     | Pre-treatment                     | Best sowing            | Depth                 | Germ. %* | Seed / 100 plants* |
|----------------------------------------------------------------------|------|----------------|-----------------------------------|------------------------|-----------------------|----------|--------------------|
| <b><i>Gompholobium tomentosum</i></b><br>Hairy Yellow Pea            | 45   | Easy-Moderate  | Dry heat or hot water             | Mar – May              | 3–4 mm                | 34%      | 0.8 g              |
| <b><i>Grevillea vestita</i></b>                                      | 47   | Moderate       | Nick seed coat; smoke water       | Mar – May              | 3–5 mm                | –        | –                  |
| <b><i>Hovea pungens</i></b><br>Devil's Pins                          | 49   | Easy-Moderate  | Boiling water 30 s                | Mar – May              | 4–6 mm                | 41%      | 3.0 g              |
| <b><i>Kennedia prostrata</i></b><br>Scarlet Runner / Running Postman | 51   | Easy           | Hot water + smoke water           | Mar – May              | About 5 mm            | 30%      | 8.3 g              |
| <b><i>Leucophyta brownii</i></b><br>Cushion Bush                     | 53   | Easy-Moderate  | None                              | Mar – May              | Surface – light cover | –        | –                  |
| <b><i>Macrozamia fraseri</i></b><br>Sandplain Zamia                  | 55   | Moderate       | Remove fleshy coat; float test    | Feb – Mar (fresh seed) | Surface               | –        | –                  |
| <b><i>Melaleuca huegelii</i></b><br>Chenille Honeymyrtle             | 57   | Easy           | None                              | Mar – Apr, Sep – Oct   | Surface               | 58%      | 0.1 g              |
| <b><i>Melaleuca systema</i></b><br>Coastal Honeymyrtle               | 59   | Easy-Moderate  | None                              | Mar – Apr, Sep – Oct   | Surface               | 14%      | 0.3 g              |
| <b><i>Myoporum insulare</i></b><br>Blueberry Tree / Common Boobialla | 61   | Moderate       | Ferment berries; smoke-water soak | Mar – May              | 2–3 mm                | –        | –                  |
| <b><i>Olearia axillaris</i></b><br>Coastal Daisybush                 | 63   | Easy-Moderate  | Air-dry 3 months; smoke water     | Aug – Sep              | Surface               | 21%      | 1.4 g              |
| <b><i>Petrophile linearis</i></b><br>Pixie Mops                      | 65   | Difficult      | Remove hairs, nick coat; smoke    | Mar – May              | 3–5 mm                | <1%†     | –                  |
| <b><i>Petrophile macrostachya</i></b>                                | 67   | Difficult *    | Remove hairs, nick coat; smoke    | Mar – May              | 3–5 mm                | –        | –                  |
| <b><i>Podotheca gnaphalioides</i></b><br>Golden Long-heads           | 69   | Moderate       | Smoke water (optional)            | Apr – May              | Surface – light cover | 7%       | 1.2 g              |
| <b><i>Ptilotus drummondii</i></b><br>Narrowleaf Mulla Mulla          | 71   | Mod.-Difficult | Remove woolly floral parts; smoke | Mar – May              | 1–2 mm                | 4%       | 8.4 g              |
| <b><i>Ptilotus manglesii</i></b><br>Pom Poms                         | 73   | Mod.-Difficult | Remove woolly floral parts; smoke | Mar – May              | 1–2 mm                | 3%       | 18 g               |
| <b><i>Ptilotus polystachyus</i></b><br>Prince of Wales Feather       | 75   | Moderate *     | Remove woolly floral parts; smoke | Mar – May              | 1–2 mm                | –        | –                  |
| <b><i>Rhagodia baccata</i></b><br>Berry Saltbush                     | 77   | Moderate       | Ferment berries; smoke-water soak | Mar – May              | 2–3 mm                | 11%      | 1.3 g              |
| <b><i>Spinifex hirsutus</i></b><br>Hairy Spinifex                    | 79   | Moderate *     | Cut off spines/awns; smoke water  | Mar – May              | 10 mm (needs dark)    | –        | –                  |
| <b><i>Spinifex longifolius</i></b><br>Beach Spinifex                 | 81   | Moderate *     | Cut off spines/awns; smoke water  | Mar – May              | 10 mm (needs dark)    | –        | –                  |
| <b><i>Spyridium globulosum</i></b><br>Basket Bush                    | 83   | Moderate       | Hot water (or dry heat)           | Apr – Jun              | 2–3 mm (needs dark)   | 41%      | 0.5 g              |
| <b><i>Threlkeldia diffusa</i></b><br>Coast Bonefruit                 | 85   | Easy-Moderate  | 24 h smoke-water soak             | Apr – Jun              | 1–2 mm                | –        | –                  |
| <b><i>Waitzia suaveolens</i></b><br>Fragrant Waitzia                 | 87   | Easy-Moderate  | Smoke water (optional)            | Apr – May              | Surface – light cover | 21%      | 0.1 g              |

\* Alcoa Marrinup Nursery routine test averages (after the pre-treatment shown). 'Seed / 100 plants' is the approximate weight of seed that produced 100 seedlings in those tests – sow extra to allow for losses. † Untreated seed – germinates poorly without the pre-treatment shown. '–' = no nursery test data.

## Sowing calendar

| Species                          | Jan | Feb  | Mar  | Apr  | May  | Jun  | Jul | Aug  | Sep  | Oct  | Nov | Dec |
|----------------------------------|-----|------|------|------|------|------|-----|------|------|------|-----|-----|
| <i>Allocasuarina fraseriana</i>  | –   | OK   | Best | Best | Best | OK   | OK  | OK   | OK   | OK   | –   | –   |
| <i>Allocasuarina humilis</i>     | –   | OK   | Best | Best | Best | OK   | OK  | OK   | OK   | OK   | –   | –   |
| <i>Allocasuarina lehmanniana</i> | –   | OK   | Best | Best | Best | OK   | OK  | OK   | OK   | OK   | –   | –   |
| <i>Anigozanthos manglesii</i>    | –   | –    | Best | Best | Best | OK   | OK  | OK   | –    | –    | –   | –   |
| <i>Atriplex isatidea</i>         | –   | –    | Best | Best | Best | OK   | OK  | OK   | OK   | OK   | –   | –   |
| <i>Austrostipa flavescens</i>    | –   | OK   | Best | Best | Best | OK   | OK  | OK   | –    | –    | –   | –   |
| <i>Banksia attenuata</i>         | –   | –    | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Banksia menziesii</i>         | –   | –    | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Banksia sessilis</i>          | –   | –    | Best | Best | Best | OK   | OK  | OK   | –    | –    | –   | –   |
| <i>Bossiaea eriocarpa</i>        | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Calothamnus quadrifidus</i>   | –   | OK   | Best | Best | OK   | OK   | OK  | OK   | Best | Best | OK  | –   |
| <i>Carpobrotus virescens</i>     | –   | OK   | Best | Best | Best | OK   | –   | –    | OK   | OK   | –   | –   |
| <i>Corymbia calophylla</i>       | –   | OK   | Best | Best | OK   | OK   | OK  | OK   | Best | Best | OK  | –   |
| <i>Daviesia divaricata</i>       | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Daviesia nudiflora</i>        | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Daviesia triflora</i>         | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Dianella revoluta</i>         | –   | OK   | Best | Best | OK   | –    | –   | –    | –    | –    | –   | –   |
| <i>Eucalyptus gomphocephala</i>  | –   | OK   | Best | Best | OK   | OK   | OK  | OK   | Best | Best | OK  | –   |
| <i>Eucalyptus marginata</i>      | –   | OK   | Best | Best | OK   | OK   | OK  | OK   | Best | Best | OK  | –   |
| <i>Gompholobium tomentosum</i>   | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Grevillea vestita</i>         | –   | –    | Best | Best | Best | OK   | OK  | OK   | –    | –    | –   | –   |
| <i>Hovea pungens</i>             | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Kennedia prostrata</i>        | –   | OK   | Best | Best | Best | OK   | OK  | OK   | Best | OK   | –   | –   |
| <i>Leucophyta brownii</i>        | –   | –    | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Macrozamia fraseri</i>        | OK  | Best | Best | OK   | OK   | –    | –   | –    | –    | –    | –   | –   |
| <i>Melaleuca huegelii</i>        | –   | OK   | Best | Best | OK   | OK   | OK  | OK   | Best | Best | OK  | –   |
| <i>Melaleuca systema</i>         | –   | OK   | Best | Best | OK   | OK   | OK  | OK   | Best | Best | OK  | –   |
| <i>Myoporum insulare</i>         | –   | –    | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Olearia axillaris</i>         | –   | –    | OK   | OK   | OK   | OK   | OK  | Best | Best | OK   | –   | –   |
| <i>Petrophile linearis</i>       | –   | –    | Best | Best | Best | OK   | OK  | OK   | –    | –    | –   | –   |
| <i>Petrophile macrostachya</i>   | –   | –    | Best | Best | Best | OK   | OK  | OK   | –    | –    | –   | –   |
| <i>Podotheca gnaphalioides</i>   | –   | –    | OK   | Best | Best | OK   | –   | –    | –    | –    | –   | –   |
| <i>Ptilotus drummondii</i>       | –   | OK   | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Ptilotus manglesii</i>        | –   | OK   | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Ptilotus polystachyus</i>     | –   | OK   | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Rhagodia baccata</i>          | –   | –    | Best | Best | Best | OK   | OK  | OK   | OK   | –    | –   | –   |
| <i>Spinifex hirsutus</i>         | OK  | OK   | Best | Best | Best | –    | –   | –    | OK   | OK   | OK  | OK  |
| <i>Spinifex longifolius</i>      | OK  | OK   | Best | Best | Best | –    | –   | –    | OK   | OK   | OK  | OK  |
| <i>Spyridium globulosum</i>      | –   | –    | OK   | Best | Best | Best | OK  | OK   | –    | –    | –   | –   |
| <i>Threlkeldia diffusa</i>       | –   | –    | OK   | Best | Best | Best | OK  | OK   | –    | –    | –   | –   |
| <i>Waitzia suaveolens</i>        | –   | –    | OK   | Best | Best | OK   | –   | –    | –    | –    | –   | –   |

Perth season

Summer

Autumn

Winter

Spring

Sum.



Best window



Workable – slower or less reliable



Avoid / low success

Work back from your planting date: most of these species need roughly 3–6 months from sowing to a plantable tube.

## Pre-treatments explained

Many WA species have seed dormancy that must be broken before they will germinate. The species pages say which treatment to use.

| Treatment            | How to do it                                                                                                                                                                                           | Used for                                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| None                 | Sow untreated seed. Remove from cones or follicles first where needed.                                                                                                                                 | <i>Allocasuarina</i> , <i>Banksia</i>                                      |
| Boiling water (30 s) | Drop seed into boiling water for 30 seconds, then drain and cool (Alcoa Marrinup method).                                                                                                              | Hard-seeded peas, e.g. <i>Daviesia</i> , <i>Hovea</i>                      |
| Hot water soak       | Pour near-boiling water over the seed (at least 10 parts water to 1 part seed) and leave to cool and soak overnight. Sow the seed that has swollen (1.5–2× its size); re-treat the rest.               | Hard-seeded peas                                                           |
| Dry heat             | Spread seed on a tray in a pre-heated oven. Peas: 80–120°C for 10 min to 1 hour. Kangaroo paw: 100°C for 3 hours.                                                                                      | <i>Gompholobium</i> , <i>Anigozanthos</i>                                  |
| Smoke water          | Soak seed for up to 24 hours in diluted smoke water (e.g. 1:9), or water trays with it after sowing. Also has slight antifungal properties.                                                            | Fire-responsive species, e.g. <i>Anigozanthos</i> (optional for others)    |
| Aged seed            | Store seed dry at room temperature for one or more years before sowing – dormancy fades with time (after-ripening).                                                                                    | <i>Anigozanthos</i>                                                        |
| Ferment              | Soak ripe fleshy fruit or berries in water for 4–5 days, wash the pulp off through a sieve, then dry the seed.                                                                                         | <i>Carpobrotus</i> , <i>Myoporum</i> , <i>Rhagodia</i>                     |
| Long soak            | Soak for 4 days, changing the water 3 times a day, to wash germination inhibitors out of the fruit.                                                                                                    | <i>Atriplex</i>                                                            |
| Clean or nick        | Remove whatever blocks germination: the woolly floral parts ( <i>Ptilotus</i> ), the fleshy coat ( <i>Macrozamia</i> ), or nick the seed coat with a scalpel ( <i>Grevillea</i> , <i>Petrophile</i> ). | <i>Ptilotus</i> , <i>Macrozamia</i> , <i>Grevillea</i> , <i>Petrophile</i> |

**Safety:** boiling water and hot ovens scald – use tongs, strainers, heat-proof containers and gloves.

## Difficulty ratings

| Rating             | Meaning                                                                              |
|--------------------|--------------------------------------------------------------------------------------|
| Easy               | Germinates reliably with the standard method; few losses.                            |
| Easy–Moderate      | Germinates well, but one factor needs care (e.g. damping off, extraction, low fill). |
| Moderate           | Needs the right pre-treatment and timing; germination may be low or variable.        |
| Moderate–Difficult | Low or erratic germination; specialist treatment usually needed.                     |
| Difficult          | Very low germination from seed; cuttings or other methods usually preferred.         |

## Confidence

| Level  | Meaning                                                                           |
|--------|-----------------------------------------------------------------------------------|
| High   | Species-specific germination data and handling guidance are published.            |
| Medium | Some species data (e.g. few tests) plus genus-level guidance. Verify with trials. |
| Low    | Little information available; treat as a trial protocol.                          |

# Standard sowing method

Species pages refer back to these steps and only note where a species differs.

| Step                         | What to do                                                                                                                                                                                                                                                       |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Seed mix                  | Sterile, free-draining, low-nutrient mix: about <b>80–85% washed river sand + 15–20% coir or peat</b> , or a commercial native seed-raising mix. Use fresh mix and clean trays. Pre-moisten before sowing.                                                       |
| 2. Check the seed            | Label every lot (species, provenance, collection date). Cut-test a few seeds – good seed is firm and white inside. For peas, a float test removes damaged seed.                                                                                                  |
| 3. Pre-treat                 | Use the treatment on the species page (see Pre-treatments explained). Sow soon after hot-water treatment.                                                                                                                                                        |
| 4. Sow                       | Cover to about <b>twice the seed's thickness</b> . Fine seed: sprinkle on the surface and press in lightly, or cover with a thin layer of fine sand or vermiculite. Large seed (e.g. <i>Banksia</i> , peas): sow direct into tubes to avoid pricking-out damage. |
| 5. Water in (post-treatment) | Water with a fine spray – smoke water is a useful first watering (slight antifungal effect, more even germination). Keep moist, never wet. A capillary bed lets you water from below.                                                                            |
| 6. Germinate                 | Shaded, sheltered, protected from rain and drying wind. Check daily. Watch for damping off, snails and slugs.                                                                                                                                                    |
| 7. Prick out                 | When seedlings can be handled – usually when the <b>second pair of leaves</b> appears; earlier for taprooted species. Lift with a narrow spatula, hold by the leaves, not the stem, and firm mix around the roots with a gentle spray.                           |
| 8. Shade house               | About <b>2 weeks</b> in the shade house after pricking out. Don't let tubes dry out (a capillary bed helps).                                                                                                                                                     |
| 9. Harden off                | Move gradually into full sun over <b>2–4 weeks</b> . Pot on to larger containers after 1–3 months if needed; don't let roots become pot-bound.                                                                                                                   |

## Seed storage

| Topic        | Guidance                                                                                                                                                                                                                                                                                                                           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General rule | Dry the seed (air-dry or desiccator, never an oven), seal it in labelled airtight containers and keep it cool and dark, away from vermin. A sealed fridge is good; freezing suits many species but not all.                                                                                                                        |
| Exceptions   | <i>Anigozanthos manglesii</i> lost germination at –18°C – store at room temperature or cooler, not frozen. <i>Allocasuarina</i> and <i>Banksia</i> seed is short-lived once out of the cone – store in the cone, or dry and cool. <i>Macrozamia fraseri</i> seed is shed moist and is damaged by cold, dry storage – sow it fresh. |
| Long-lived   | Hard-coated pea seed keeps many years dry (related <i>Acacia</i> seed has stayed viable 50+ years).                                                                                                                                                                                                                                |
| Collecting   | Collecting from bushland generally needs a DBCA licence and landholder permission. Collect only ripe seed from many plants to keep genetic diversity.                                                                                                                                                                              |

## Hygiene and damping off

Damping off (*Pythium*, *Rhizoctonia*) rots seedlings at soil level, especially in wet, crowded trays. Prevent it with sterile mix, clean trays, thin sowing, good airflow and bottom watering. Remove affected seedlings quickly and use a registered fungicide only if the problem persists. Keep pots off the ground to reduce *Phytophthora* risk.

# *Allocasuarina fraseriana*

Sheoak / Western Sheoak

Family  
Casuarinaceae

| DIFFICULTY | PRE-TREATMENT | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|------------|---------------|-------------|--------------|---------------------|
| Easy       | None          | Mar – May   | 2–3 mm       | 37%                 |

|        | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|--------|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| Sowing | –   | OK  | Best | Best | Best | OK  | OK  | OK  | OK  | OK  | –   | –   |

|             |                                    |                     |
|-------------|------------------------------------|---------------------|
| Best window | Workable – slower or less reliable | Avoid / low success |
|-------------|------------------------------------|---------------------|

## Plant snapshot

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| Habit      | Erect tree, 5–15 m. Male and female flowers usually on separate trees (dioecious).              |
| Flowering  | May – Oct. Woody cones up to 40 mm long follow on female trees.                                 |
| Habitat    | Lateritic soils and white, grey or yellow sand; jarrah forest and sand dunes.                   |
| Local note | Recorded in the Wanneroo LGA. Only female trees bear cones – note which trees you collect from. |

## Propagation options

| Method   | Suitability | Notes                        | Details in |
|----------|-------------|------------------------------|------------|
| Cuttings | Not used    | Rarely used for sheoaks.     | –          |
| Seed     | Main method | No pre-treatment; 2–5 weeks. | This guide |
| Division | Not used    | Tree.                        | –          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Cones mature within about a year of flowering. Collect cones that have turned pale to dark brown <b>with the valves still closed</b> . Use secateurs or a pole pruner. |
| Extraction & cleaning | Place cones in a paper bag or tray in a warm, dry spot – they open readily and drop the winged seeds (samaras). Sieve off cone debris.                                 |
| Seed quality          | Many seeds look sound but are empty. <b>Cut-test</b> a sample before sowing and discard lots with empty or shrivelled seed.                                            |
| Storage life          | <b>Short once released from the cone</b> unless kept dry and cool. Store dried seed sealed in the fridge, or keep seed in closed cones until the sowing season.        |
| Typical germination   | About <b>37%</b> (untreated, Alcoa nursery average). About <b>76 seedlings per gram</b> – roughly 1.3 g of seed per 100 seedlings.                                     |

## Sowing & growing on

| Item             | Details                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time | <b>Mar – May</b> (cooling autumn weather, 15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan (heat).                                                  |
| Seed mix         | Standard sterile native seed mix (see Standard method).                                                                                                      |
| Pre-treatment    | <b>None needed</b> . Seed becomes sticky (mucilaginous) when wet – mix dry seed with a little dry, sterile yellow sand so it spreads evenly and can be seen. |

| Item                               | Details                                                                                                                                                                                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sowing depth &amp; method</b>   | Scatter thinly and cover lightly with about <b>2–3 mm</b> of fine sand or vermiculite. <b>Direct sowing into tubes or cells</b> (a few seeds each, thinned later) reduces losses – sheoak seedlings are prone to damping off after pricking out. |
| <b>After sowing</b>                | Water in with a fine spray (smoke water is optional – aerosol smoke gave similar results to untreated seed in Alcoa trials). Keep moist, not wet. Watch for damping off.                                                                         |
| <b>Germination time</b>            | Seedlings emerge after about <b>2–5 weeks</b> .                                                                                                                                                                                                  |
| <b>Pricking out</b>                | If sown in trays, prick out early – once seedlings are about 2–3 cm tall and can be handled – disturbing roots as little as possible.                                                                                                            |
| <b>Shade house &amp; hardening</b> | About <b>2 weeks</b> in the shade house after pricking out or thinning, then move gradually to full sun over 2–4 weeks.                                                                                                                          |
| <b>Time to planting size</b>       | Est. 4–6 months from sowing (record your own).                                                                                                                                                                                                   |

### Common problems

| Problem                                 | Likely cause and fix                                                                              |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Seedlings collapse at soil level</b> | Damping off. Sow thinly, use sterile mix, water from below where possible, direct sow into tubes. |
| <b>Very low germination</b>             | Seed lot mostly empty. Cut-test before sowing; collect from closed, mature cones.                 |
| <b>Seed clumps together</b>             | Mucilage. Mix dry seed with dry sand before sowing; don't pre-soak.                               |

### Notes

|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| <b>Other notes</b> | Cuttings are rarely used. Seed from several trees keeps genetic diversity in revegetation stock. |
| <b>Confidence</b>  | <b>High.</b> Published nursery test data and WA seed-handling notes are available.               |
| <b>Sources</b>     | See References: 1, 2, 3, 4, 5                                                                    |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# *Allocasuarina humilis*

Dwarf Sheoak

Family  
Casuarinaceae

| DIFFICULTY    | PRE-TREATMENT | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|---------------|---------------|-------------|--------------|---------------------|
| Easy-Moderate | None          | Mar – May   | 2–3 mm       | 25%                 |

|        | Jan         | Feb                                | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct                 | Nov | Dec |
|--------|-------------|------------------------------------|------|------|------|-----|-----|-----|-----|---------------------|-----|-----|
| Sowing | –           | OK                                 | Best | Best | Best | OK  | OK  | OK  | OK  | OK                  | –   | –   |
|        | Best window | Workable – slower or less reliable |      |      |      |     |     |     |     | Avoid / low success |     |     |

## Plant snapshot

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| Habit      | Erect or spreading shrub, 0.2–2 m. Dioecious or monoecious.                               |
| Flowering  | May – Nov.                                                                                |
| Habitat    | Sand (often over laterite), sandy clay and gravel on sandplains and dunes.                |
| Local note | Widespread in the South West. Collect from local-provenance plants and record the source. |

## Propagation options

| Method   | Suitability | Notes                             | Details in |
|----------|-------------|-----------------------------------|------------|
| Cuttings | Not used    | Rarely used for sheoaks.          | –          |
| Seed     | Main method | No pre-treatment; sow generously. | This guide |
| Division | Not used    | Woody shrub.                      | –          |

## Seed collection & storage

| Item                  | Details                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect mature brown cones <b>with valves still closed</b> , usually within a year of flowering. On dioecious plants only female plants bear cones. |
| Extraction & cleaning | Dry cones in a paper bag in a warm, dry place until they open; sieve out the small winged seeds.                                                    |
| Seed quality          | Germination is often low – <b>cut-test</b> a sample. Many seeds can be empty.                                                                       |
| Storage life          | <b>Short once released from the cone</b> unless kept dry and cool. Store sealed in the fridge, or keep in closed cones until needed.                |
| Typical germination   | About <b>25%</b> (untreated, Alcoa nursery average). About <b>77 seedlings per gram</b> – roughly 1.3 g of seed per 100 seedlings.                  |

## Sowing & growing on

| Item                  | Details                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> (15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan.                                                                                                                  |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                             |
| Pre-treatment         | <b>None needed.</b> Mix dry seed with dry, sterile sand for even spreading (seed turns sticky when wet).                                                                                      |
| Sowing depth & method | Cover lightly, about <b>2–3 mm</b> . Direct sow a few seeds per tube/cell and thin, or sow thinly in trays. Sow more generously than for <i>A. fraseriana</i> to allow for lower germination. |
| After sowing          | Water in with a fine spray; keep moist, not wet. Good airflow to limit damping off.                                                                                                           |
| Germination time      | About <b>2–5 weeks</b> .                                                                                                                                                                      |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | Prick out early, at about 2–3 cm tall, with minimal root disturbance.                                       |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem                          | Likely cause and fix                                           |
|----------------------------------|----------------------------------------------------------------|
| Seedlings collapse at soil level | Damping off. Sow thinly, sterile mix, avoid overwatering.      |
| Very low germination             | Empty seed. Cut-test; collect from closed, fully mature cones. |
| Seed clumps together             | Mix dry seed with dry sand; don't pre-soak.                    |

### Notes

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| Other notes | Slower-growing shrub – allow extra time to reach planting size.                    |
| Confidence  | <b>High.</b> Published nursery test data and WA seed-handling notes are available. |
| Sources     | See References: 1, 2, 3, 4, 5                                                      |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# *Allocasuarina lehmanniana*

Dune Sheoak

Family  
Casuarinaceae

| DIFFICULTY    | PRE-TREATMENT | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|---------------|---------------|-------------|--------------|---------------------|
| Easy-Moderate | None          | Mar – May   | 2–3 mm       | 19%                 |

|        | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|--------|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| Sowing | –   | OK  | Best | Best | Best | OK  | OK  | OK  | OK  | OK  | –   | –   |

|             |                                    |                     |
|-------------|------------------------------------|---------------------|
| Best window | Workable – slower or less reliable | Avoid / low success |
|-------------|------------------------------------|---------------------|

## Plant snapshot

|            |                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|
| Habit      | Shrub, 0.5–4 m. Male and female flowers on separate plants (dioecious).                                 |
| Flowering  | Sep – Dec (female flowers red).                                                                         |
| Habitat    | Coastal sites and winter-wet areas.                                                                     |
| Local note | Recorded in Joondalup and Wanneroo. Only female plants bear cones – note which plants you collect from. |

## Propagation options

| Method   | Suitability | Notes                                 | Details in                 |
|----------|-------------|---------------------------------------|----------------------------|
| Cuttings | Not used    | Rarely used for sheoaks.              | –                          |
| Seed     | Main method | No pre-treatment; like other sheoaks. | <a href="#">This guide</a> |
| Division | Not used    | Woody shrub.                          | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect mature brown cones <b>with the valves still closed</b> , usually within a year of flowering. Use secateurs.                          |
| Extraction & cleaning | Dry cones in a paper bag in a warm, dry spot until they open; sieve out the small winged seeds.                                              |
| Seed quality          | Many seeds can be empty – <b>cut-test</b> a sample before sowing.                                                                            |
| Storage life          | <b>Short once released from the cone</b> unless kept dry and cool. Store sealed in the fridge, or keep seed in closed cones until needed.    |
| Typical germination   | About <b>19%</b> (untreated, Alcoa nursery average, 3 tests). About <b>193 seedlings per gram</b> – roughly 0.5 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> (15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan.                                   |
| Seed mix              | Standard sterile native seed mix.                                                                              |
| Pre-treatment         | <b>None needed.</b> Mix dry seed with dry, sterile sand for even spreading (seed turns sticky when wet).       |
| Sowing depth & method | Cover lightly, about <b>2–3 mm</b> . Direct sow a few seeds per tube or cell and thin, or sow thinly in trays. |
| After sowing          | Water in with a fine spray; keep moist, not wet. Good airflow to limit damping off.                            |
| Germination time      | About <b>2–5 weeks</b> .                                                                                       |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | Prick out early, at about 2–3 cm tall, with minimal root disturbance.                                       |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem                          | Likely cause and fix                                           |
|----------------------------------|----------------------------------------------------------------|
| Seedlings collapse at soil level | Damping off. Sow thinly, sterile mix, avoid overwatering.      |
| Very low germination             | Empty seed. Cut-test; collect from closed, fully mature cones. |
| Seed clumps together             | Mix dry seed with dry sand; don't pre-soak.                    |

### Notes

|             |                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| Other notes | A good choice for damp, low-lying coastal sites. Collect from several female plants to keep genetic diversity.    |
| Confidence  | <b>Medium.</b> Nursery germination data (3 tests); handling follows the other <i>Allocasuarina</i> in this guide. |
| Sources     | See References: 2, 3, 4                                                                                           |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Anigozanthos manglesii

Red and Green Kangaroo Paw

Family  
Haemodoraceae

| DIFFICULTY | PRE-TREATMENT                    | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|------------|----------------------------------|-------------|--------------|---------------------|
| Moderate   | Dry heat 100°C 3 h, or aged seed | Mar – May   | 1–2 mm       | 70%                 |

|        | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|--------|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| Sowing | –   | –   | Best | Best | Best | OK  | OK  | OK  | –   | –   | –   | –   |

Best window
  Workable – slower or less reliable
  Avoid / low success

## Plant snapshot

|            |                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------|
| Habit      | Rhizomatous perennial herb, 0.2–1.1 m (flower stems). Often short-lived – best treated as a 2–3 year plant. |
| Flowering  | Aug – Nov (green and red). WA's floral emblem.                                                              |
| Habitat    | White, yellow or grey sand and sandy loam.                                                                  |
| Local note | Recorded in the Wanneroo LGA. Also called Mangles Kangaroo Paw.                                             |

## Propagation options

| Method   | Suitability | Notes                                                            | Details in                 |
|----------|-------------|------------------------------------------------------------------|----------------------------|
| Cuttings | Not used    | Clumping herb – no stems to cut.                                 | –                          |
| Seed     | Main method | Dry heat or aged seed; autumn sowing.                            | <a href="#">This guide</a> |
| Division | Good        | Split vigorous clumps into rhizome pieces with shoots and roots. | → Division Guide           |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Seed ripens roughly three months after flowering (about Nov – Feb). Collect when capsules dry and brown, before they split and shed.                                                                                                   |
| Extraction & cleaning | Dry capsules in paper bags; rub and sieve to separate the small seed.                                                                                                                                                                  |
| Seed quality          | Fresh seed is strongly dormant – untreated germination can be very low (about 1% in one Alcoa trial).                                                                                                                                  |
| Storage life          | <b>Stores well for several years</b> dry at room temperature, and germination <b>improves with age</b> (after-ripening). Research found <b>freezing (–18°C) reduced germination</b> – store dry at 23°C or cooler, not in the freezer. |
| Typical germination   | About <b>70%</b> after heat treatment (Alcoa nursery average). About <b>211 seedlings per gram</b> – roughly 0.5 g of seed per 100 seedlings.                                                                                          |

## Sowing & growing on

| Item             | Details                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------|
| Best sowing time | <b>Mar – May</b> – germinates in cool conditions (tested at 15–20°C). Jun – Aug workable (slower). Avoid Sep – Feb. |
| Seed mix         | Standard sterile native seed mix; very free-draining.                                                               |

| Item                               | Details                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-treatment</b>               | Choose one: <ul style="list-style-type: none"> <li>• <b>Dry heat:</b> spread seed on a tray in an oven at <b>100°C for 3 hours</b> (Geraldton community nursery; Alcoa).</li> <li>• <b>Aged seed:</b> use seed stored at room temperature for a few years.</li> <li>• <b>Smoke water</b> (e.g. 24 h soak at 1:9) – response improves the longer seed has been stored.</li> </ul> |
| <b>Sowing depth &amp; method</b>   | Sow thinly on the surface of trays and cover very lightly, about <b>1–2 mm</b> of fine sand or vermiculite.                                                                                                                                                                                                                                                                      |
| <b>After sowing</b>                | Water in with a fine spray (smoke water optional). Keep moist but not wet. Protect from snails and slugs.                                                                                                                                                                                                                                                                        |
| <b>Germination time</b>            | About <b>3–6 weeks</b> (around 20 days in the Geraldton nursery).                                                                                                                                                                                                                                                                                                                |
| <b>Pricking out</b>                | When the <b>second leaf</b> appears and seedlings can be handled (about 2–3 cm). Lift gently with a narrow spatula.                                                                                                                                                                                                                                                              |
| <b>Shade house &amp; hardening</b> | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. Keep foliage dry and airflow good.                                                                                                                                                                                                                                   |
| <b>Time to planting size</b>       | Est. 4–6 months from sowing (record your own).                                                                                                                                                                                                                                                                                                                                   |

### Common problems

| Problem                                | Likely cause and fix                                                                  |
|----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Almost no germination</b>           | Seed still dormant. Heat-treat, use older seed, or add smoke water.                   |
| <b>Black spots spreading on leaves</b> | Ink disease (fungal). Remove affected plants, improve airflow, avoid wetting foliage. |
| <b>Seedlings chewed</b>                | Snails and slugs. Bait or hand-remove.                                                |

### Notes

|                    |                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------|
| <b>Other notes</b> | Division of clumps also works. Seed-grown plants vary slightly; keep local provenance for revegetation. |
| <b>Confidence</b>  | <b>High.</b> Well researched: dormancy, heat, smoke and storage responses are published.                |
| <b>Sources</b>     | See References: 1, 3, 5, 6, 7, 8, 9                                                                     |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Atriplex isatidea

Coast Saltbush

Family  
Amaranthaceae  
(chenopods)

| DIFFICULTY    |  | PRE-TREATMENT                      |     | BEST SOWING |      | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |     |     |     |
|---------------|--|------------------------------------|-----|-------------|------|--------------|-----|---------------------|-----|-----|-----|-----|-----|
| Easy-Moderate |  | Soak 4 days; smoke water           |     | Mar – May   |      | 1–2 mm       |     | 22%                 |     |     |     |     |     |
|               |  | Jan                                | Feb | Mar         | Apr  | May          | Jun | Jul                 | Aug | Sep | Oct | Nov | Dec |
| Sowing        |  | –                                  | –   | Best        | Best | Best         | OK  | OK                  | OK  | OK  | OK  | –   | –   |
| Best window   |  | Workable – slower or less reliable |     |             |      |              |     | Avoid / low success |     |     |     |     |     |

Plant snapshot

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| Habit      | Grey-leaved, spreading shrub, 0.5–2.5 m. Male and female flowers on the same plant. |
| Flowering  | Mainly Mar – Jun.                                                                   |
| Habitat    | White sand on coastal dunes.                                                        |
| Local note | Recorded in the Wanneroo LGA. Very salt- and wind-tolerant.                         |

Propagation options

| Method   | Suitability | Notes                                                                    | Details in       |
|----------|-------------|--------------------------------------------------------------------------|------------------|
| Cuttings | Good        | Saltbush cuttings are generally reliable (provisional for this species). | → Cuttings Guide |
| Seed     | Main method | Soak 4 days in changed water, then smoke water.                          | This guide       |
| Division | Not used    | Woody shrub.                                                             | –                |

Seed collection & storage

| Item                  | Details                                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Each seed is enclosed in a pair of papery bracts. Collect the fruiting bracts when they turn dry and straw-coloured and come away easily. |
| Extraction & cleaning | Rub off stems and leaves and sieve. Sow the seed still inside its bracts.                                                                 |
| Seed quality          | Soaking or washing removes germination inhibitors (salts) from the bracts – the reason for the long soak below.                           |
| Storage life          | Not well documented. Store dry and cool. Soaked seed can be dried again and stored until needed (Geraldton community nursery).            |
| Typical germination   | About 22% (untreated, Alcoa nursery average, 3 tests). About 15 seedlings per gram – roughly 6.5 g of seed per 100 seedlings.             |

Sowing & growing on

| Item                  | Details                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – May. Jun – Oct is workable (autumn, early winter and spring suit saltbush). Avoid Nov – Feb.                                      |
| Seed mix              | Standard sterile native seed mix.                                                                                                       |
| Pre-treatment         | Soak for 4 days, changing the water 3 times a day, then water in with smoke water (Geraldton method). Presoaked seed germinates faster. |
| Sowing depth & method | Cover about 1–2 mm. Sow thinly in trays, or 2–3 bracts per tube and thin.                                                               |
| After sowing          | Water in with smoke water; keep moist, not wet.                                                                                         |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Germination time        | About <b>11–20 days</b> (Geraldton nursery). Germinates best in cool conditions.                            |
| Pricking out            | When the second pair of leaves appears.                                                                     |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem                    | Likely cause and fix                                                        |
|----------------------------|-----------------------------------------------------------------------------|
| Poor, patchy germination   | Inhibitors not washed out. Soak the full 4 days with regular water changes. |
| Seedlings collapse         | Damping off. Thinner sowing, sterile mix, good airflow.                     |
| Leaf spots on young plants | Foliage kept wet. Water the mix, not the leaves; improve airflow.           |

### Notes

|             |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------|
| Other notes | Tip cuttings are a workable alternative (see the Cuttings Guide).                                  |
| Confidence  | <b>Medium.</b> Nursery germination data (3 tests) and a WA community-nursery method are available. |
| Sources     | See References: 3, 4, 5, 33, 38                                                                    |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Austrostipa flavescens

Coast Speargrass

Family  
Poaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT          |     |      |      | BEST SOWING |     |     | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |  |
|-----------------------------------------------------------------------------------------------------|------------------------|-----|------|------|-------------|-----|-----|--------------|-----|---------------------|-----|-----|--|
| Easy-Moderate                                                                                       | Smoke water; aged seed |     |      |      | Mar – May   |     |     | 0–2 mm       |     | Not recorded        |     |     |  |
|                                                                                                     | Jan                    | Feb | Mar  | Apr  | May         | Jun | Jul | Aug          | Sep | Oct                 | Nov | Dec |  |
| Sowing                                                                                              | –                      | OK  | Best | Best | Best        | OK  | OK  | OK           | –   | –                   | –   | –   |  |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                        |     |      |      |             |     |     |              |     |                     |     |     |  |

## Plant snapshot

|            |                                                                              |
|------------|------------------------------------------------------------------------------|
| Habit      | Tufted perennial grass, 0.2–1.2 m.                                           |
| Flowering  | Sep – Oct.                                                                   |
| Habitat    | Grey, red or calcareous sand, sandy loam and limestone; sand dunes.          |
| Local note | Coastal species. Collect from local-provenance stands and record the source. |

## Propagation options

| Method   | Suitability | Notes                                          | Details in                 |
|----------|-------------|------------------------------------------------|----------------------------|
| Cuttings | Not used    | Tufted grass.                                  | –                          |
| Seed     | Main method | Aged seed and smoke water; sow shallow.        | <a href="#">This guide</a> |
| Division | Good        | Divide established tussocks into small clumps. | → Division Guide           |

## Seed collection & storage

| Item                  | Details                                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Seed ripens in late spring to early summer after flowering. Collect when seed heads turn straw-coloured and seed pulls away easily; it sheds quickly once ripe. |
| Extraction & cleaning | Strip seed by hand and dry in paper bags. Long awns can be trimmed to make sowing easier.                                                                       |
| Seed quality          | Fresh seed is often dormant (after-ripening needed).                                                                                                            |
| Storage life          | Store dry at room temperature. <b>Older seed (1–2 years) gives higher germination</b> in <i>Austrostipa</i> (Geraldton community nursery).                      |
| Typical germination   | No Alcoa data for this species. A related speargrass ( <i>A. caespitosa</i> ) gave 73% after aerosol smoke.                                                     |

## Sowing & growing on

| Item                  | Details                                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> (autumn). Feb and Jun – Aug workable. Avoid Sep – Jan.                                                                                                                                                                                             |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                                                                                                   |
| Pre-treatment         | <ul style="list-style-type: none"> <li>• Use seed stored for a year or more, <b>or</b></li> <li>• Water trays with smoke water after sowing (Geraldton method).</li> <li>• With fresh seed only, gibberellic acid plus smoke water can lift germination.</li> </ul> |
| Sowing depth & method | <b>Sow very shallowly</b> – on or just below the surface (0–2 mm), pressed in lightly. Seed needs light to germinate.                                                                                                                                               |
| After sowing          | Water in with smoke water; keep moist.                                                                                                                                                                                                                              |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Germination time        | About <b>21 days</b> (Geraldton nursery).                                                                   |
| Pricking out            | Prick out or divide small clumps when seedlings are 3–5 cm with several leaves.                             |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem                          | Likely cause and fix                                                  |
|----------------------------------|-----------------------------------------------------------------------|
| Poor germination from fresh seed | Dormant. Store seed a year, or use smoke water (and GA if available). |
| Seed buried too deep             | Needs light. Sow on the surface and press in.                         |
| Seedlings rot                    | Overwatering. Keep moist, not wet.                                    |

### Notes

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| Other notes | Can also be multiplied by dividing established clumps.                                                        |
| Confidence  | <b>Medium.</b> WA community-nursery method and genus-level dormancy research; no Alcoa data for this species. |
| Sources     | See References: 3, 5, 23                                                                                      |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Banksia attenuata

Slender Banksia / Candle Banksia

Family  
Proteaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT             |     |      |      | BEST SOWING |     | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |     |
|-----------------------------------------------------------------------------------------------------|---------------------------|-----|------|------|-------------|-----|--------------|-----|---------------------|-----|-----|-----|
| Easy-Moderate                                                                                       | None (extract from cones) |     |      |      | Mar – May   |     | 3–5 mm       |     | 60%                 |     |     |     |
|                                                                                                     | Jan                       | Feb | Mar  | Apr  | May         | Jun | Jul          | Aug | Sep                 | Oct | Nov | Dec |
| Sowing                                                                                              | –                         | –   | Best | Best | Best        | OK  | OK           | OK  | OK                  | –   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                           |     |      |      |             |     |              |     |                     |     |     |     |

## Plant snapshot

|            |                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|
| Habit      | Lignotuberous tree or shrub, usually 2–4 m (up to 10 m). Resprouts after fire.                          |
| Flowering  | Oct – Feb (yellow candles).                                                                             |
| Habitat    | White, yellow, brown or pale red sand, sometimes over laterite; sand dunes and sandplains.              |
| Local note | Recorded in Joondalup and Wanneroo. Highly susceptible to <i>Phytophthora</i> dieback – strict hygiene. |

## Propagation options

| Method   | Suitability | Notes                                    | Details in                 |
|----------|-------------|------------------------------------------|----------------------------|
| Cuttings | Difficult   | Banksias are rarely grown from cuttings. | –                          |
| Seed     | Main method | Extract from cones; no pre-treatment.    | <a href="#">This guide</a> |
| Division | Not used    | Tree / shrub.                            | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Follicles can take a year or more to mature. Collect <b>old, dry, dark grey-brown cones</b> with closed follicles. Cones hold seed for some time, so collection isn't tied to one month.                                                                                              |
| Extraction & cleaning | Open follicles with heat: oven at <b>120–130°C for 1 hour</b> (or 200°C for 10 min), or briefly heat then soak in water for several hours and dry in a warm oven. Separate the winged black seed from the woody separator. Good seed is white and firm inside. Never microwave cones. |
| Seed quality          | Seed from older cones tends to have lower viability than from younger cones – favour recent cones.                                                                                                                                                                                    |
| Storage life          | <b>Short-lived once extracted</b> unless stored dry and cool. Best kept in the cone until the sowing season; extracted seed can remain viable for some years sealed, dry and cool (10–25°C, 40–60% RH).                                                                               |
| Typical germination   | About <b>60%</b> (untreated, Alcoa nursery average). About <b>15 seedlings per gram</b> – roughly 7 g of seed per 100 seedlings.                                                                                                                                                      |

## Sowing & growing on

| Item             | Details                                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time | <b>Mar – May</b> – sow once the weather cools after March. Jun – Sep workable. Avoid Oct – Feb (heat raises damping-off losses).   |
| Seed mix         | Free-draining, <b>pathogen-free, low-phosphorus</b> mix (pasteurise home-made mix at 60°C for 30 min). Slightly acid (about pH 6). |
| Pre-treatment    | <b>None needed</b> once seed is out of the follicle.                                                                               |

| Item                    | Details                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sowing depth & method   | <b>Direct sow 1–2 seeds per deep tube</b> (preferred), or sow thinly in deep trays. <b>Just cover</b> with fine mix or vermiculite (about 3–5 mm).                                                                         |
| After sowing            | Water in with a fine spray. This species is <b>highly vulnerable to damping off</b> – water from below (capillary bed) if possible, or use a registered fungicide at germination (e.g. furalaxyl or captan, as per label). |
| Germination time        | About <b>16–49 days</b> .                                                                                                                                                                                                  |
| Pricking out            | If tray-sown, prick out <b>at the cotyledon stage, no later than a few weeks after germination</b> , before the taproot lengthens.                                                                                         |
| Shade house & hardening | About <b>2 weeks</b> under shade cloth after pricking out, then harden off gradually to full sun over 2–4 weeks.                                                                                                           |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                                                                                                                                             |

### Common problems

| Problem                                | Likely cause and fix                                                                              |
|----------------------------------------|---------------------------------------------------------------------------------------------------|
| Seedlings collapse soon after emerging | Damping off. Sterile mix, bottom watering, fungicide at germination, less crowding.               |
| Yellowing, dying seedlings in pots     | Phosphorus toxicity or <i>Phytophthora</i> . Use very low-P fertiliser; keep pots off the ground. |
| Few seeds from cones                   | Immature or old cones. Collect ripe, recent cones; check follicles have developed.                |

### Notes

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| Other notes | Feed only with half-strength, very low-phosphorus fertiliser once established (Proteaceae are P-sensitive). |
| Confidence  | <b>High.</b> Published nursery test data and Banksia seed-handling guidance are available.                  |
| Sources     | See References: 1, 3, 10, 11, 12, 13                                                                        |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
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|           |                       |           |          |        |         |       |

# Banksia menziesii

Firewood Banksia

Family  
Proteaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT             |     |      |      | BEST SOWING |     | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |     |
|-----------------------------------------------------------------------------------------------------|---------------------------|-----|------|------|-------------|-----|--------------|-----|---------------------|-----|-----|-----|
| Easy-Moderate                                                                                       | None (extract from cones) |     |      |      | Mar – May   |     | 3–5 mm       |     | 56%                 |     |     |     |
|                                                                                                     | Jan                       | Feb | Mar  | Apr  | May         | Jun | Jul          | Aug | Sep                 | Oct | Nov | Dec |
| Sowing                                                                                              | –                         | –   | Best | Best | Best        | OK  | OK           | OK  | OK                  | –   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                           |     |      |      |             |     |              |     |                     |     |     |     |

## Plant snapshot

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| Habit      | Tree or shrub, 1.3–7 m, from a lignotuber. Resprouts after fire.                                  |
| Flowering  | Feb – Oct, peaking May – Jul (pink, red or yellow).                                               |
| Habitat    | White, grey or yellow sand.                                                                       |
| Local note | Recorded in Joondalup and Wanneroo. Medium to high susceptibility to <i>Phytophthora</i> dieback. |

## Propagation options

| Method   | Suitability | Notes                                    | Details in                 |
|----------|-------------|------------------------------------------|----------------------------|
| Cuttings | Difficult   | Banksias are rarely grown from cuttings. | –                          |
| Seed     | Main method | Extract from cones; no pre-treatment.    | <a href="#">This guide</a> |
| Division | Not used    | Tree / shrub.                            | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Up to 25 follicles per cone, each with 1–2 seeds. Follicles open mainly after fire (or release seed after about two years), so collect <b>mature, closed cones</b> .                                |
| Extraction & cleaning | Heat cones to open follicles: oven at <b>120–130°C for 1 hour</b> (or 200°C for 10 min), or briefly heat, soak for several hours, then dry in a warm oven. Separate seed from the woody separators. |
| Seed quality          | Good seed is white and firm with a dark coat. Recent cones generally give better viability than old cones.                                                                                          |
| Storage life          | <b>Short-lived once extracted</b> unless stored dry and cool. Keep in the cone until needed where possible.                                                                                         |
| Typical germination   | About <b>56%</b> (untreated, Alcoa nursery average). About <b>27 seedlings per gram</b> – roughly 3.7 g of seed per 100 seedlings.                                                                  |

## Sowing & growing on

| Item                  | Details                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> , once the weather cools. Jun – Sep workable. Avoid Oct – Feb.                            |
| Seed mix              | Free-draining, pathogen-free, <b>low-phosphorus</b> mix (about pH 6).                                      |
| Pre-treatment         | <b>None needed</b> once seed is out of the follicle.                                                       |
| Sowing depth & method | <b>Direct sow 1–2 seeds per deep tube</b> (preferred), or thinly in deep trays. Just cover (about 3–5 mm). |

| Item                    | Details                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| After sowing            | Water in with a fine spray; bottom watering where possible. Watch for damping off; fungicide at germination if needed (as per label). |
| Germination time        | About <b>26–40 days</b> .                                                                                                             |
| Pricking out            | If tray-sown, prick out at the cotyledon stage within a few weeks of germination.                                                     |
| Shade house & hardening | About <b>2 weeks</b> under shade cloth after pricking out, then harden off gradually to full sun over 2–4 weeks.                      |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                                                        |

### Common problems

| Problem                                | Likely cause and fix                                                                               |
|----------------------------------------|----------------------------------------------------------------------------------------------------|
| Seedlings collapse soon after emerging | Damping off. Sterile mix, bottom watering, less crowding.                                          |
| Yellowing or dying in pots             | Phosphorus toxicity or <i>Phytophthora</i> . Very low-P feed; strict hygiene; pots off the ground. |
| Follicles won't open                   | Not enough heat. Repeat heating, or heat then soak and re-dry.                                     |

### Notes

|             |                                                                                            |
|-------------|--------------------------------------------------------------------------------------------|
| Other notes | Takes about five to seven years to flower from seed.                                       |
| Confidence  | <b>High.</b> Published nursery test data and Banksia seed-handling guidance are available. |
| Sources     | See References: 1, 3, 10, 11, 12, 13                                                       |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# *Banksia sessilis*

Parrot Bush

Family  
Proteaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT               |     |      | BEST SOWING |      |     | SOWING DEPTH |     |     | TYPICAL GERMINATION |     |     |
|-----------------------------------------------------------------------------------------------------|-----------------------------|-----|------|-------------|------|-----|--------------|-----|-----|---------------------|-----|-----|
| Easy-Moderate                                                                                       | None (smoke water optional) |     |      | Mar – May   |      |     | 3–5 mm       |     |     | 42%                 |     |     |
|                                                                                                     | Jan                         | Feb | Mar  | Apr         | May  | Jun | Jul          | Aug | Sep | Oct                 | Nov | Dec |
| Sowing                                                                                              | –                           | –   | Best | Best        | Best | OK  | OK           | OK  | –   | –                   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                             |     |      |             |      |     |              |     |     |                     |     |     |

## Plant snapshot

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| Habit      | Upright, prickly shrub or small tree to 6 m. Killed by fire and regrows from seed.        |
| Flowering  | Mainly Jul – Nov (cream-yellow domes).                                                    |
| Habitat    | Sandy soils over laterite or limestone.                                                   |
| Local note | The Swan Coastal Plain form is var. <i>cygnorum</i> . Formerly <i>Dryandra sessilis</i> . |

## Propagation options

| Method   | Suitability | Notes                                          | Details in                 |
|----------|-------------|------------------------------------------------|----------------------------|
| Cuttings | Not used    | Cuttings have proved virtually impossible.     | –                          |
| Seed     | Main method | No pre-treatment; sow as it cools after March. | <a href="#">This guide</a> |
| Division | Not used    | Shrub.                                         | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Not serotinous: most seed is <b>released in autumn</b> even without fire. Collect ripe heads in summer, before release.                                |
| Extraction & cleaning | Put heads in a paper bag in a warm place or a low oven ( <b>no hotter than 60°C</b> ) until the follicles open; cut stubborn follicles with secateurs. |
| Seed quality          | Good seed is firm and white inside.                                                                                                                    |
| Storage life          | Seed in the soil is short-lived, but cleaned seed keeps well <b>stored dry and cool</b> .                                                              |
| Typical germination   | About <b>42%</b> (untreated, Alcoa nursery average). About <b>33 seedlings per gram</b> – roughly 3 g of seed per 100 seedlings.                       |

## Sowing & growing on

| Item                  | Details                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> – sow once the weather cools after March. Jun – Aug workable. Avoid Sep – Feb. |
| Seed mix              | Free-draining, sterile, <b>low-phosphorus</b> mix.                                              |
| Pre-treatment         | <b>None needed.</b> Geraldton soaks seed in smoke water (1:9) for 24 hours.                     |
| Sowing depth & method | Direct sow 1–2 seeds per tube, just covered (about 3–5 mm).                                     |
| After sowing          | Water in; avoid excess water – damping off can be a problem.                                    |
| Germination time      | About <b>2–6 weeks</b> (14 days at Geraldton).                                                  |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | If tray-sown, prick out at the cotyledon stage, within a few weeks of germination.                          |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem             | Likely cause and fix                                                     |
|---------------------|--------------------------------------------------------------------------|
| Seedlings collapse  | Damping off. Water less, sterile mix, good airflow.                      |
| Few seeds in heads  | Collected too late – seed already released in autumn. Collect in summer. |
| Yellowing seedlings | Phosphorus toxicity. Use very low-P fertiliser.                          |

### Notes

|             |                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------|
| Other notes | Cuttings have proved virtually impossible – seed is the only practical method. Flowers within about two years. |
| Confidence  | <b>High.</b> Nursery test data (as <i>Dryandra sessilis</i> ) and WA seed-handling notes available.            |
| Sources     | See References: 3, 5, 20, 25                                                                                   |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Bossiaea eriocarpa

Common Brown Pea

Family  
Fabaceae

| DIFFICULTY    | PRE-TREATMENT      | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|---------------|--------------------|-------------|--------------|---------------------|
| Easy-Moderate | Boiling water 30 s | Mar – May   | 2–3 mm       | 44%                 |

|                        | Jan | Feb                                           | Mar  | Apr  | May  | Jun | Jul | Aug | Sep  | Oct                            | Nov | Dec |
|------------------------|-----|-----------------------------------------------|------|------|------|-----|-----|-----|------|--------------------------------|-----|-----|
| Sowing                 | –   | OK                                            | Best | Best | Best | OK  | OK  | OK  | Best | OK                             | –   | –   |
| <div>Best window</div> |     | <div>Workable – slower or less reliable</div> |      |      |      |     |     |     |      | <div>Avoid / low success</div> |     |     |

Plant snapshot

|            |                                               |
|------------|-----------------------------------------------|
| Habit      | Erect or straggly, spreading shrub, 0.2–1 m.  |
| Flowering  | Jul – Nov (yellow and brown-red pea flowers). |
| Habitat    | Usually sandy soils.                          |
| Local note | Recorded in Joondalup and Wanneroo.           |

Propagation options

| Method   | Suitability | Notes               | Details in |
|----------|-------------|---------------------|------------|
| Cuttings | Not used    | Not documented.     | –          |
| Seed     | Main method | Boiling water 30 s. | This guide |
| Division | Not used    | –                   | –          |

Seed collection & storage

| Item                  | Details                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods ripen in summer and shed seed when ripe. <b>Bag developing pods</b> with stockings or muslin.                                           |
| Extraction & cleaning | Rub and sieve the small seed from pod fragments.                                                                                             |
| Seed quality          | Hard-seeded – fresh seed is dormant until the coat is treated. Float test to remove damaged seed.                                            |
| Storage life          | <b>Long.</b> Keeps many years when dry (typical of hard-coated pea seed).                                                                    |
| Typical germination   | About <b>44%</b> after boiling-water treatment (Alcoa nursery average). About <b>229 seedlings per gram</b> – under 0.5 g per 100 seedlings. |

Sowing & growing on

| Item                  | Details                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> and <b>Sep.</b> Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.                                                           |
| Seed mix              | Standard sterile native seed mix.                                                                                                            |
| Pre-treatment         | <b>Boiling water for 30 seconds</b> (Alcoa method), or pour near-boiling water over and soak overnight. Sow swollen seed; re-treat the rest. |
| Sowing depth & method | Small seed – cover about <b>2–3 mm</b> . Direct sow a few seeds per tube and thin.                                                           |
| After sowing          | Water in; keep moist, not wet.                                                                                                               |
| Germination time      | Can start about <b>one week</b> after treatment.                                                                                             |
| Pricking out          | At the cotyledon to first true leaf stage, before the taproot lengthens.                                                                     |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem            | Likely cause and fix                               |
|--------------------|----------------------------------------------------|
| Low germination    | Seed coat not broken. Re-treat unswollen seed.     |
| Swollen seed rots  | Over-boiled or too wet. Keep to 30 s; water less.  |
| Seedlings collapse | Damping off. Sterile mix, airflow, avoid crowding. |

### Notes

|             |                                                                                       |
|-------------|---------------------------------------------------------------------------------------|
| Other notes | Very small seed for a pea – a little goes a long way.                                 |
| Confidence  | <b>High.</b> Nursery germination data (40 tests) and WA hard-seed guidance available. |
| Sources     | See References: 3, 5, 14, 15, 16                                                      |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Calothamnus quadrifidus

One-sided Bottlebrush

Family  
Myrtaceae

| DIFFICULTY | PRE-TREATMENT | BEST SOWING          | SOWING DEPTH | TYPICAL GERMINATION |
|------------|---------------|----------------------|--------------|---------------------|
| Easy       | None          | Mar – Apr, Sep – Oct | Surface      | 26%                 |

|        | Jan | Feb | Mar  | Apr  | May | Jun | Jul | Aug | Sep  | Oct  | Nov | Dec |
|--------|-----|-----|------|------|-----|-----|-----|-----|------|------|-----|-----|
| Sowing | –   | OK  | Best | Best | OK  | OK  | OK  | OK  | Best | Best | OK  | –   |

|                                                                                               |                                                                                                                      |                                                                                                         |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|  Best window |  Workable – slower or less reliable |  Avoid / low success |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

## Plant snapshot

|            |                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------|
| Habit      | Erect, compact or spreading shrub, usually 0.9–2 m.                                                |
| Flowering  | Jun – Dec (red, one-sided brushes).                                                                |
| Habitat    | Wide variety of soils and habitats.                                                                |
| Local note | Recorded in the Wanneroo LGA. Several subspecies exist – label stock by subspecies and provenance. |

## Propagation options

| Method   | Suitability | Notes                                                     | Details in       |
|----------|-------------|-----------------------------------------------------------|------------------|
| Cuttings | Good        | Presumed to strike like related melaleucas (provisional). | → Cuttings Guide |
| Seed     | Main method | Easy from seed held in old capsules.                      | This guide       |
| Division | Not used    | Woody shrub.                                              | –                |

## Seed collection & storage

| Item                  | Details                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Woody capsules stay on the older stems for years. Collect capsules from <b>older wood</b> (at least a year old) at any time of year.  |
| Extraction & cleaning | Place capsules in an open tray or paper bag in a warm spot; fine seed is released in a few days. No need to separate seed from chaff. |
| Seed quality          | Very fine seed mixed with chaff – a cut-test and count gives seeds per gram.                                                          |
| Storage life          | <b>Long</b> if kept dry and cool (typical of capsule-fruited Myrtaceae).                                                              |
| Typical germination   | About <b>26%</b> (untreated, Alcoa nursery average). About <b>572 seedlings per gram</b> – a pinch goes a long way.                   |

## Sowing & growing on

| Item                  | Details                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – Apr</b> and <b>Sep – Oct</b> (15–25°C). Feb, May – Aug and Nov workable. Avoid Dec – Jan.                                |
| Seed mix              | Fine-textured, sterile native seed mix with a smooth surface.                                                                     |
| Pre-treatment         | <b>None needed.</b>                                                                                                               |
| Sowing depth & method | Mix seed with dry, sterile yellow sand and sprinkle from a spice jar. <b>Sow on the surface</b> and press in lightly; don't bury. |
| After sowing          | Water in with smoke water (Geraldton method) using a very fine spray, or water from below (bog method) so seed isn't washed away. |
| Germination time      | About <b>7–10 days</b> (Geraldton nursery).                                                                                       |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | When the <b>second pair of leaves</b> appears (about 2 cm). Seedlings are tiny – handle by the leaves.      |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem                 | Likely cause and fix                                              |
|-------------------------|-------------------------------------------------------------------|
| Seedlings too crowded   | Sown too thickly. Dilute seed well with sand.                     |
| Seed washed into clumps | Overhead watering too coarse. Use a fine mist or bottom watering. |
| Damping off             | Crowding and wet mix. Thin sowing, airflow, sterile mix.          |

### Notes

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| Other notes | Cuttings are worth trialling if a particular form is wanted (see the Cuttings Guide).     |
| Confidence  | <b>High.</b> Large nursery data set (39 tests) and WA community-nursery method available. |
| Sources     | See References: 3, 4, 5, 18                                                               |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Carpobrotus virescens

Coastal Pigface

Family  
Aizoaceae

| DIFFICULTY    | PRE-TREATMENT              |     |      |      | BEST SOWING |     | SOWING DEPTH |     |     | TYPICAL GERMINATION |     |     |
|---------------|----------------------------|-----|------|------|-------------|-----|--------------|-----|-----|---------------------|-----|-----|
| Easy-Moderate | Ferment fruit; smoke water |     |      |      | Mar – May   |     | Surface      |     |     | 21%                 |     |     |
|               | Jan                        | Feb | Mar  | Apr  | May         | Jun | Jul          | Aug | Sep | Oct                 | Nov | Dec |
| Sowing        | –                          | OK  | Best | Best | Best        | OK  | –            | –   | OK  | OK                  | –   | –   |
| Best window   |                            |     |      |      |             |     |              |     |     |                     |     |     |
|               |                            |     |      |      |             |     |              |     |     |                     |     |     |

## Plant snapshot

|            |                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Habit      | Prostrate succulent groundcover, 0.1–0.3 m high and 0.5–3 m wide.                                                                                                 |
| Flowering  | Jun – Dec or Jan (purple-pink, sometimes white), followed by fleshy fruit.                                                                                        |
| Habitat    | Coastal limestone and sand dunes.                                                                                                                                 |
| Local note | Recorded in the Wanneroo LGA. Collect only from plants you are sure are native – the introduced Hottentot Fig ( <i>C. edulis</i> ) is naturalised on some coasts. |

## Propagation options

| Method   | Suitability | Notes                                                       | Details in       |
|----------|-------------|-------------------------------------------------------------|------------------|
| Cuttings | Main method | Stem pieces root very easily; rooted runners can be lifted. | → Cuttings Guide |
| Seed     | Good        | Ferment fruit, sow fine seed on the surface; 6–8 days.      | This guide       |
| Division | Possible    | Lift stems that have rooted – treat as cuttings.            | –                |

## Seed collection & storage

| Item                  | Details                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect the fleshy fruits when soft and fully ripe, after flowering.                                                                                         |
| Extraction & cleaning | <b>Ferment:</b> soak the crushed fruit in water for 4–5 days, then wash the tiny seeds out through a fine sieve and dry them.                                |
| Seed quality          | Seed is very fine – each fruit holds many seeds.                                                                                                             |
| Storage life          | Not well documented. Store dry and cool.                                                                                                                     |
| Typical germination   | About <b>21%</b> after a dry-heat treatment (Alcoa nursery average, 9 tests). About <b>454 seedlings per gram</b> – roughly 0.2 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May.</b> Feb, Jun and Sep – Oct are workable. Avoid the coldest and hottest months.                                                      |
| Seed mix              | Standard sterile native seed mix, extra free-draining.                                                                                            |
| Pre-treatment         | Ferment and wash the seed (above). Alcoa used a dry-heat pulse instead. Water in with smoke water after sowing.                                   |
| Sowing depth & method | <b>Surface-sow.</b> Mix the fine seed with dry yellow sand in a spice jar and shake it evenly over the tray (Geraldton method). Press in lightly. |
| After sowing          | Water in with smoke water using a fine mist; keep just moist – succulent seedlings rot if wet.                                                    |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Germination time        | About <b>6–8 days</b> (Geraldton nursery).                                                                  |
| Pricking out            | When seedlings have two pairs of leaves and can be handled.                                                 |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem                 | Likely cause and fix                                                                   |
|-------------------------|----------------------------------------------------------------------------------------|
| Seedlings rot           | Too wet. Water less, improve drainage and airflow.                                     |
| Seed washed into clumps | Use bottom watering or a fine mist.                                                    |
| Poor germination        | Fruit not fully ripe, or seed not cleaned. Collect soft, ripe fruit and ferment fully. |

### Notes

|             |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------|
| Other notes | Stem cuttings are quicker and very reliable (see the Cuttings Guide).                              |
| Confidence  | <b>Medium.</b> Nursery germination data (9 tests) and a WA community-nursery method are available. |
| Sources     | See References: 3, 4, 5, 32                                                                        |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Corymbia calophylla

Marri

Family  
Myrtaceae

| DIFFICULTY | PRE-TREATMENT | BEST SOWING          | SOWING DEPTH | TYPICAL GERMINATION |
|------------|---------------|----------------------|--------------|---------------------|
| Easy       | None          | Mar – Apr, Sep – Oct | About 5 mm   | 60%                 |

|        | Jan | Feb | Mar  | Apr  | May | Jun | Jul | Aug | Sep  | Oct  | Nov | Dec |
|--------|-----|-----|------|------|-----|-----|-----|-----|------|------|-----|-----|
| Sowing | –   | OK  | Best | Best | OK  | OK  | OK  | OK  | Best | Best | OK  | –   |

|             |                                    |                     |
|-------------|------------------------------------|---------------------|
| Best window | Workable – slower or less reliable | Avoid / low success |
|-------------|------------------------------------|---------------------|

## Plant snapshot

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| Habit      | Tree to 40 m (rarely a mallee), with rough, tessellated bark.                              |
| Flowering  | Dec or Jan – May (white or pink).                                                          |
| Habitat    | Wide range: clay loam, sandy clay, gravel, grey sand over limestone, granite and laterite. |
| Local note | Recorded in the Wanneroo LGA. Large urn-shaped gumnuts ('honky nuts').                     |

## Propagation options

| Method   | Suitability | Notes                                       | Details in |
|----------|-------------|---------------------------------------------|------------|
| Cuttings | Not used    | Eucalypts are very difficult from cuttings. | –          |
| Seed     | Main method | No pre-treatment; about 60% germination.    | This guide |
| Division | Not used    | Tree.                                       | –          |

## Seed collection & storage

| Item                  | Details                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Fruit takes about a year to mature. Collect brown, woody gumnuts when the valves look ready to open.               |
| Extraction & cleaning | Keep gumnuts warm and dry in a tray or paper bag until the large winged seed is released.                          |
| Seed quality          | Large seed – easy to check by cutting a few open.                                                                  |
| Storage life          | Long if stored cool and dry (eucalypt seed keeps its viability a long time).                                       |
| Typical germination   | About 60% (untreated, Alcoa nursery average). About 19 seedlings per gram – roughly 5 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan.                      |
| Seed mix              | Standard sterile native seed mix.                                                               |
| Pre-treatment         | None needed. A dip in dilute bleach (10%) to surface-sterilise the seed can help.               |
| Sowing depth & method | Large seed – cover about 5 mm. Direct sow 1–2 seeds per tube (preferred) or space out in trays. |
| After sowing          | Water in; keep moist.                                                                           |
| Germination time      | Usually within about 1–2 weeks.                                                                 |
| Pricking out          | When seedlings have two pairs of true leaves.                                                   |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem            | Likely cause and fix                                |
|--------------------|-----------------------------------------------------|
| Seedlings collapse | Damping off. Sterile mix, avoid crowding.           |
| Leggy seedlings    | Too much shade. Move to brighter light once up.     |
| Roots circling     | Kept too long in tubes. Plant out or pot on sooner. |

### Notes

|             |                                                                                      |
|-------------|--------------------------------------------------------------------------------------|
| Other notes | Fast-growing – time sowings so trees aren't held in tubes too long.                  |
| Confidence  | <b>High.</b> Large nursery data set (66 tests) and WA eucalypt seed notes available. |
| Sources     | See References: 3, 4, 5, 19                                                          |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Daviesia divaricata

Marno

Family  
Fabaceae

|            |                    |             |              |                     |
|------------|--------------------|-------------|--------------|---------------------|
| DIFFICULTY | PRE-TREATMENT      | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
| Moderate   | Boiling water 30 s | Mar – May   | 3–5 mm       | 28%                 |

|        |             |                                    |      |      |      |     |     |     |      |                     |     |     |
|--------|-------------|------------------------------------|------|------|------|-----|-----|-----|------|---------------------|-----|-----|
|        | Jan         | Feb                                | Mar  | Apr  | May  | Jun | Jul | Aug | Sep  | Oct                 | Nov | Dec |
| Sowing | –           | OK                                 | Best | Best | Best | OK  | OK  | OK  | Best | OK                  | –   | –   |
|        | Best window | Workable – slower or less reliable |      |      |      |     |     |     |      | Avoid / low success |     |     |

Plant snapshot

|            |                                     |
|------------|-------------------------------------|
| Habit      | Shrub, 0.4–2 m.                     |
| Flowering  | Aug – Nov (pea flowers).            |
| Habitat    | Sand, limestone and laterite.       |
| Local note | Recorded in Joondalup and Wanneroo. |

Propagation options

| Method   | Suitability | Notes               | Details in |
|----------|-------------|---------------------|------------|
| Cuttings | Not used    | Not documented.     | –          |
| Seed     | Main method | Boiling water 30 s. | This guide |
| Division | Not used    | –                   | –          |

Seed collection & storage

| Item                  | Details                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods ripen in late spring–summer and <b>release seed as soon as they are ripe</b> . Tie fine mesh or stocking bags over developing pods and collect once they split. |
| Extraction & cleaning | Rub pods and sieve or winnow to remove debris.                                                                                                                       |
| Seed quality          | Hard-seeded: fresh seed stays dormant until the coat is treated. Float test: discard seed that floats.                                                               |
| Storage life          | <b>Long</b> . Hard-coated pea seed keeps for many years when dry.                                                                                                    |
| Typical germination   | About <b>28%</b> after boiling-water treatment (Alcoa nursery average, 12 tests). About <b>11 seedlings per gram</b> – roughly 9 g of seed per 100 seedlings.        |

Sowing & growing on

| Item                  | Details                                                                                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> and <b>Sep</b> . Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.                                                                                                                           |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                                                 |
| Pre-treatment         | <b>Boiling water</b> : drop seed into boiling water for <b>30 seconds</b> (Alcoa method), then cool; or pour near-boiling water over the seed (10:1) and soak overnight. Sow the swollen seed; re-treat the rest. |
| Sowing depth & method | Cover about <b>3–5 mm</b> . Direct sow 2–3 seeds per tube and thin, or sow in trays.                                                                                                                              |
| After sowing          | Water in; keep moist, not wet. Don't let swollen seed dry out.                                                                                                                                                    |
| Germination time      | Can start about <b>one week</b> after treatment.                                                                                                                                                                  |
| Pricking out          | At the cotyledon to first true leaf stage – early, before the taproot lengthens.                                                                                                                                  |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem              | Likely cause and fix                                       |
|----------------------|------------------------------------------------------------|
| Very low germination | Seed coat not broken. Re-treat unswollen seed.             |
| Swollen seed rots    | Mix too wet, or boiled too long. Keep to 30 s; water less. |
| Seedlings collapse   | Damping off. Sterile mix, good airflow.                    |

### Notes

|             |                                                                                            |
|-------------|--------------------------------------------------------------------------------------------|
| Other notes | Boiling water scalds – use tongs, a strainer and heat-proof containers.                    |
| Confidence  | <b>High.</b> Nursery germination data from 12 tests; method follows WA hard-seed guidance. |
| Sources     | See References: 3, 4, 14, 15                                                               |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Daviesia nudiflora

No widely used common name (a bitter-pea)

Family  
**Fabaceae**

| DIFFICULTY | PRE-TREATMENT      | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|------------|--------------------|-------------|--------------|---------------------|
| Moderate   | Boiling water 30 s | Mar – May   | 3–5 mm       | 22%                 |

|        | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep  | Oct | Nov | Dec |
|--------|-----|-----|------|------|------|-----|-----|-----|------|-----|-----|-----|
| Sowing | –   | OK  | Best | Best | Best | OK  | OK  | OK  | Best | OK  | –   | –   |

Best window

Workable – slower or less reliable

Avoid / low success

### Plant snapshot

|            |                                                           |
|------------|-----------------------------------------------------------|
| Habit      | Shrub, 0.3–1.5 m.                                         |
| Flowering  | Jun – Sep (orange-yellow and red-brown pea flowers).      |
| Habitat    | Sandy and clayey soils and laterite; low rises and flats. |
| Local note | Recorded in the Wanneroo LGA.                             |

### Propagation options

| Method   | Suitability | Notes               | Details in |
|----------|-------------|---------------------|------------|
| Cuttings | Not used    | Not documented.     | –          |
| Seed     | Main method | Boiling water 30 s. | This guide |
| Division | Not used    | –                   | –          |

### Seed collection & storage

| Item                  | Details                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods ripen in summer and <b>release seed as soon as they are ripe</b> . Tie nylon stockings or muslin bags over developing pods and collect once they split.     |
| Extraction & cleaning | Rub pods and sieve or winnow to remove debris.                                                                                                                   |
| Seed quality          | Hard-seeded: nearly all fresh seed is dormant until the seed coat is treated. Float test: discard seed that floats.                                              |
| Storage life          | <b>Long</b> . Hard-coated pea-family seed keeps for many years when dry (related <i>Acacia</i> seed has stayed viable 50+ years).                                |
| Typical germination   | About <b>22%</b> after boiling-water treatment (Alcoa nursery average, few tests). About <b>57 seedlings per gram</b> – roughly 1.7 g of seed per 100 seedlings. |

### Sowing & growing on

| Item                  | Details                                                                                                                                                                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> and <b>Sep</b> . Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan (heat stress, damping off).                                                                                                                                                          |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                                                                                                           |
| Pre-treatment         | <b>Boiling water:</b> drop seed into boiling water for <b>30 seconds</b> (Alcoa method), then cool; or pour near-boiling water over seed (at least 10 parts water to 1 part seed) and leave to soak overnight. <b>Sow seed that has swollen;</b> re-treat seed that hasn't. |
| Sowing depth & method | Cover to about twice the seed's thickness, roughly <b>3–5 mm</b> . Direct sow 2–3 seeds per tube and thin, or sow in trays.                                                                                                                                                 |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| After sowing            | Water in; keep moist, not wet. Sow soon after treatment – don't let swollen seed dry out.                   |
| Germination time        | Can start about <b>one week</b> after treatment.                                                            |
| Pricking out            | At the cotyledon to first true leaf stage – early, before the taproot lengthens.                            |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem              | Likely cause and fix                                                                              |
|----------------------|---------------------------------------------------------------------------------------------------|
| Very low germination | Seed coat not broken. Re-treat unswollen seed; try a dry-heat pulse (80–120°C) on the next batch. |
| Swollen seed rots    | Mix too wet or seed damaged by overlong boiling. Keep boiling to 30 s; water less.                |
| Seedlings collapse   | Damping off. Sterile mix, good airflow, avoid crowding.                                           |

### Notes

|             |                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------|
| Other notes | Treat with care: boiling water scalds – use tongs, a strainer and heat-proof containers.                         |
| Confidence  | <b>Medium.</b> Nursery germination data available (small number of tests); method follows WA hard-seed guidance. |
| Sources     | See References: 1, 3, 5, 14, 15, 16                                                                              |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

Family  
**Fabaceae**

No widely used common name (a bitter-pea)

|             | Jan                                | Feb | Mar  | Apr  | May  | Jun | Jul | Aug                 | Sep  | Oct | Nov | Dec |
|-------------|------------------------------------|-----|------|------|------|-----|-----|---------------------|------|-----|-----|-----|
| Sowing      | –                                  | OK  | Best | Best | Best | OK  | OK  | OK                  | Best | OK  | –   | –   |
| Best window | Workable – slower or less reliable |     |      |      |      |     |     | Avoid / low success |      |     |     |     |

|                   |                                                |
|-------------------|------------------------------------------------|
| <b>Habit</b>      | Broom-like, almost leafless shrub, 0.25–0.6 m. |
| <b>Flowering</b>  | May – Sep (pea flowers).                       |
| <b>Habitat</b>    | White or grey sand and laterite.               |
| <b>Local note</b> | Recorded in the Wanneroo LGA.                  |

| Method   | Suitability | Notes               | Details in                 |
|----------|-------------|---------------------|----------------------------|
| Cuttings | Not used    | Not documented.     | –                          |
| Seed     | Main method | Boiling water 30 s. | <a href="#">This guide</a> |
| Division | Not used    | –                   | –                          |

| Item                  | Details                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods ripen in late spring–summer and shed seed as soon as ripe. Bag developing pods with fine mesh and collect once they split.                              |
| Extraction & cleaning | Rub pods and sieve or winnow.                                                                                                                                |
| Seed quality          | Hard-seeded. Float test: discard seed that floats.                                                                                                           |
| Storage life          | <b>Long.</b> Many years when dry.                                                                                                                            |
| Typical germination   | About <b>26%</b> after boiling-water treatment (Alcoa nursery average, 2 tests). About <b>11 seedlings per gram</b> – roughly 9 g of seed per 100 seedlings. |

| Item                  | Details                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> and <b>Sep.</b> Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan.                                         |
| Seed mix              | Standard sterile native seed mix.                                                                                              |
| Pre-treatment         | <b>Boiling water for 30 seconds</b> (Alcoa method), or a near-boiling overnight soak. Sow the swollen seed; re-treat the rest. |
| Sowing depth & method | Cover about <b>3–5 mm</b> . Direct sow 2–3 seeds per tube and thin.                                                            |
| After sowing          | Water in; keep moist, not wet.                                                                                                 |
| Germination time      | Can start about <b>one week</b> after treatment.                                                                               |
| Pricking out          | Early, at the cotyledon to first true leaf stage.                                                                              |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem              | Likely cause and fix                           |
|----------------------|------------------------------------------------|
| Very low germination | Seed coat not broken. Re-treat unswollen seed. |
| Swollen seed rots    | Too wet or over-boiled.                        |
| Seedlings collapse   | Damping off. Sterile mix, good airflow.        |

### Notes

|             |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------|
| Other notes | Small, slender plant – grow in deep tubes and avoid overpotting.                                   |
| Confidence  | <b>Medium.</b> Nursery germination data available (2 tests); method follows WA hard-seed guidance. |
| Sources     | See References: 3, 4, 14, 15                                                                       |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Dianella revoluta

Blueberry Lily / Flax Lily

Family  
Asphodelaceae  
(Hemerocallidoideae)

|            |                               |             |              |                     |
|------------|-------------------------------|-------------|--------------|---------------------|
| DIFFICULTY | PRE-TREATMENT                 | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
| Difficult  | GA 24 h + smoke<br>water 24 h | Mar – Apr   | 3–5 mm       | <1% untreated       |

|             |     |     |      |      |     |     |     |     |     |     |     |     |
|-------------|-----|-----|------|------|-----|-----|-----|-----|-----|-----|-----|-----|
|             | Jan | Feb | Mar  | Apr  | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
| Sowing      | –   | OK  | Best | Best | OK  | –   | –   | –   | –   | –   | –   | –   |
| Best window |     |     |      |      |     |     |     |     |     |     |     |     |
|             |     |     |      |      |     |     |     |     |     |     |     |     |
|             |     |     |      |      |     |     |     |     |     |     |     |     |

Plant snapshot

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| Habit      | Rhizomatous, tufted perennial herb, 0.3–1.5 m.                                      |
| Flowering  | Sep – Dec (blue-purple), followed by blue berries.                                  |
| Habitat    | White or yellow sand, sandy clay, laterite, loam and clay.                          |
| Local note | Recorded in Joondalup and Wanneroo (var. <i>divaricata</i> is common on the coast). |

Propagation options

| Method   | Suitability | Notes                               | Details in       |
|----------|-------------|-------------------------------------|------------------|
| Cuttings | Not used    | Clumping herb.                      | –                |
| Seed     | Difficult   | GA + smoke water; 8–11 weeks.       | This guide       |
| Division | Main method | The quickest way to produce plants. | → Division Guide |

Seed collection & storage

| Item                  | Details                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect ripe, deep-blue berries in summer after flowering.                                                                |
| Extraction & cleaning | Soak berries in water for a few days, then rub and wash off the pulp in a sieve. Dry the shiny black seed before storing. |
| Seed quality          | Seed is dormant (morphophysiological dormancy): untreated seed gave under 1% in Alcoa tests.                              |
| Storage life          | Not well documented. Store cleaned, dried seed sealed and cool.                                                           |
| Typical germination   | Untreated: <b>under 1%</b> (Alcoa, 19 tests). Treated seed does much better – record your own results.                    |

Sowing & growing on

| Item                  | Details                                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – Apr – seed experiences warm conditions, then germinates as the weather cools (Geraldton: not until late April). Feb and May workable.                                             |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                       |
| Pre-treatment         | Gibberellic acid (1 part per thousand) for 24 hours, then smoke water (1:9) for 24 hours (Geraldton method). Research shows warm stratification followed by smoke also breaks dormancy. |
| Sowing depth & method | Cover about 3–5 mm. Sow in trays or direct into tubes.                                                                                                                                  |
| After sowing          | Keep moist and be patient – don't discard trays too early.                                                                                                                              |
| Germination time      | About 58–75 days (Geraldton nursery).                                                                                                                                                   |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | When seedlings have 2-3 leaves and are large enough to handle.                                              |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2-4 weeks. |
| Time to planting size   | Est. 5-7 months from sowing (record your own).                                                              |

### Common problems

| Problem                       | Likely cause and fix                                                         |
|-------------------------------|------------------------------------------------------------------------------|
| Nothing after 6 weeks         | Normal – germination takes 8-11 weeks. Keep trays moist.                     |
| No germination at all         | Seed still dormant. Use GA + smoke water, or keep trays for a second season. |
| Pulp left on seed goes mouldy | Clean seed thoroughly before drying and storing.                             |

### Notes

|             |                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------|
| Other notes | Division of established clumps is a quicker way to produce plants.                                       |
| Confidence  | <b>Medium.</b> WA community-nursery method and dormancy research available; nursery results vary widely. |
| Sources     | See References: 3, 4, 5, 23                                                                              |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Eucalyptus gomphocephala

Tuart

Family  
Myrtaceae

| DIFFICULTY | PRE-TREATMENT | BEST SOWING          | SOWING DEPTH | TYPICAL GERMINATION |
|------------|---------------|----------------------|--------------|---------------------|
| Easy       | None          | Mar – Apr, Sep – Oct | 1–2 mm       | 42%                 |

|        | Jan         | Feb                                | Mar  | Apr  | May | Jun | Jul | Aug | Sep  | Oct  | Nov | Dec                 |
|--------|-------------|------------------------------------|------|------|-----|-----|-----|-----|------|------|-----|---------------------|
| Sowing | –           | OK                                 | Best | Best | OK  | OK  | OK  | OK  | Best | Best | OK  | –                   |
|        | Best window | Workable – slower or less reliable |      |      |     |     |     |     |      |      |     | Avoid / low success |

Plant snapshot

|            |                                           |
|------------|-------------------------------------------|
| Habit      | Tree, 10–40 m.                            |
| Flowering  | Jan – Apr (white).                        |
| Habitat    | Sand over limestone on the coastal plain. |
| Local note | Recorded in Joondalup and Wanneroo.       |

Propagation options

| Method   | Suitability | Notes                                       | Details in |
|----------|-------------|---------------------------------------------|------------|
| Cuttings | Not used    | Eucalypts are very difficult from cuttings. | –          |
| Seed     | Main method | No pre-treatment; 1–2 weeks.                | This guide |
| Division | Not used    | Tree.                                       | –          |

Seed collection & storage

| Item                  | Details                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Fruit takes about a year to mature. Collect brown capsules when the valves are just starting to open; many trees hold seed for a year. |
| Extraction & cleaning | Keep capsules warm and dry from a day to several weeks until seed drops. Seed is mixed with chaff.                                     |
| Seed quality          | A cut-test and count on seed and chaff estimates seeds per gram.                                                                       |
| Storage life          | Long if stored cool and dry.                                                                                                           |
| Typical germination   | About 42% (untreated, Alcoa nursery average). About 228 seedlings per gram – under 0.5 g per 100 seedlings.                            |

Sowing & growing on

| Item                  | Details                                                                    |
|-----------------------|----------------------------------------------------------------------------|
| Best sowing time      | Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan. |
| Seed mix              | Standard sterile native seed mix.                                          |
| Pre-treatment         | None needed. Surface-sterilising in dilute bleach (10%) can help.          |
| Sowing depth & method | Mix with dry sand and sprinkle thinly; cover very lightly (about 1–2 mm).  |
| After sowing          | Water in with a fine spray (smoke water optional); keep moist.             |
| Germination time      | About 1–2 weeks.                                                           |
| Pricking out          | When seedlings have two pairs of true leaves.                              |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem            | Likely cause and fix                              |
|--------------------|---------------------------------------------------|
| Seedlings collapse | Damping off. Thin sowing, sterile mix, airflow.   |
| Uneven germination | Chaff clumps. Mix seed well with sand.            |
| Roots circling     | Held too long in tubes. Time sowings to planting. |

### Notes

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Other notes | Local coastal-plain provenance matters for revegetation – record seed source. |
| Confidence  | <b>High.</b> Nursery data (16 tests) and WA eucalypt seed notes available.    |
| Sources     | See References: 3, 4, 5, 19                                                   |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
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|           |                       |           |          |        |         |       |

# Eucalyptus marginata

Jarrah

Family  
Myrtaceae

|            |               |                      |              |                     |
|------------|---------------|----------------------|--------------|---------------------|
| DIFFICULTY | PRE-TREATMENT | BEST SOWING          | SOWING DEPTH | TYPICAL GERMINATION |
| Easy       | None          | Mar – Apr, Sep – Oct | 2–3 mm       | 31%                 |

|        |             |                                    |      |      |     |     |     |     |      |                     |     |     |
|--------|-------------|------------------------------------|------|------|-----|-----|-----|-----|------|---------------------|-----|-----|
|        | Jan         | Feb                                | Mar  | Apr  | May | Jun | Jul | Aug | Sep  | Oct                 | Nov | Dec |
| Sowing | –           | OK                                 | Best | Best | OK  | OK  | OK  | OK  | Best | Best                | OK  | –   |
|        | Best window | Workable – slower or less reliable |      |      |     |     |     |     |      | Avoid / low success |     |     |

Plant snapshot

|            |                                                              |
|------------|--------------------------------------------------------------|
| Habit      | Tree to 40 m.                                                |
| Flowering  | Jun – Dec or Jan (white-cream to pink).                      |
| Habitat    | Grey sand, clay or sandy loam and laterite; hills and rises. |
| Local note | Recorded in Joondalup and Wanneroo.                          |

Propagation options

|          |             |                                             |            |
|----------|-------------|---------------------------------------------|------------|
| Method   | Suitability | Notes                                       | Details in |
| Cuttings | Not used    | Eucalypts are very difficult from cuttings. | –          |
| Seed     | Main method | No pre-treatment; 1–2 weeks.                | This guide |
| Division | Not used    | Tree.                                       | –          |

Seed collection & storage

|                       |                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Item                  | Details                                                                                                            |
| Seed collection       | Fruit takes about a year to mature. Collect brown capsules when the valves look ready to open.                     |
| Extraction & cleaning | Keep capsules warm and dry until seed is released; sieve out large debris.                                         |
| Seed quality          | Cut-test a sample; seed is mixed with chaff.                                                                       |
| Storage life          | Long if stored cool and dry.                                                                                       |
| Typical germination   | About 31% (untreated, Alcoa nursery average). About 33 seedlings per gram – roughly 3 g of seed per 100 seedlings. |

Sowing & growing on

|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| Item                  | Details                                                                                        |
| Best sowing time      | Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan.                     |
| Seed mix              | Standard sterile native seed mix.                                                              |
| Pre-treatment         | None needed. Surface-sterilising in dilute bleach (10%) can help.                              |
| Sowing depth & method | Cover lightly, about 2–3 mm. Direct sow a few seeds per tube and thin, or sow thinly in trays. |
| After sowing          | Water in; keep moist.                                                                          |
| Germination time      | About 1–2 weeks.                                                                               |
| Pricking out          | When seedlings have two pairs of true leaves.                                                  |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem            | Likely cause and fix                                     |
|--------------------|----------------------------------------------------------|
| Seedlings collapse | Damping off. Sterile mix, avoid crowding.                |
| Low germination    | Old or immature seed. Collect mature capsules; cut-test. |
| Roots circling     | Held too long in tubes.                                  |

### Notes

|             |                                                                                      |
|-------------|--------------------------------------------------------------------------------------|
| Other notes | Jarrah seedlings develop a lignotuber early – a good sign of healthy stock.          |
| Confidence  | <b>High.</b> Large nursery data set (67 tests) and WA eucalypt seed notes available. |
| Sources     | See References: 3, 4, 5, 19                                                          |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Gompholobium tomentosum

Hairy Yellow Pea

Family  
Fabaceae

|               |                       |             |              |                     |
|---------------|-----------------------|-------------|--------------|---------------------|
| DIFFICULTY    | PRE-TREATMENT         | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
| Easy-Moderate | Dry heat or hot water | Mar – May   | 3–4 mm       | 34%                 |

|                        |     |                                               |      |      |      |     |     |     |      |                                |     |     |
|------------------------|-----|-----------------------------------------------|------|------|------|-----|-----|-----|------|--------------------------------|-----|-----|
|                        | Jan | Feb                                           | Mar  | Apr  | May  | Jun | Jul | Aug | Sep  | Oct                            | Nov | Dec |
| Sowing                 | –   | OK                                            | Best | Best | Best | OK  | OK  | OK  | Best | OK                             | –   | –   |
| <div>Best window</div> |     | <div>Workable – slower or less reliable</div> |      |      |      |     |     |     |      | <div>Avoid / low success</div> |     |     |

Plant snapshot

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| Habit      | Erect, often spindly shrub, 0.15–1 m.                                                           |
| Flowering  | Jul – Jan (yellow pea flowers).                                                                 |
| Habitat    | White, grey or yellow sand, sandy clay and lateritic sand; coastal limestone, dunes and plains. |
| Local note | Recorded in Joondalup and Wanneroo.                                                             |

Propagation options

| Method   | Suitability | Notes                  | Details in |
|----------|-------------|------------------------|------------|
| Cuttings | Not used    | Not documented.        | –          |
| Seed     | Main method | Dry heat or hot water. | This guide |
| Division | Not used    | –                      | –          |

Seed collection & storage

| Item                  | Details                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods ripen in summer and shed seed when ripe. <b>Bag developing pods</b> with stockings or muslin and collect once split.                     |
| Extraction & cleaning | Rub and sieve out the small seed.                                                                                                             |
| Seed quality          | Hard-seeded – almost all fresh seed is dormant until the coat is treated.                                                                     |
| Storage life          | <b>Long.</b> Keeps for many years when dry (typical of hard-coated pea-family seed).                                                          |
| Typical germination   | About <b>34%</b> after heat treatment (Alcoa nursery average). About <b>125 seedlings per gram</b> – roughly 0.8 g of seed per 100 seedlings. |

Sowing & growing on

| Item                  | Details                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> and <b>Sep.</b> Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.                                                                                                                                                                                                                                                       |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                                                                                                                                                                        |
| Pre-treatment         | Either: <ul style="list-style-type: none"><li>• <b>Dry heat shock:</b> oven at 80–120°C for 10 min to 1 hour (Alcoa used a dry-heat pulse for this species); or</li><li>• <b>Hot water:</b> pour near-boiling water over the seed and let it cool in the water (easiest method). Sow seed that has swollen; re-treat the rest.</li></ul> |
| Sowing depth & method | Small seed – cover about <b>3–4 mm</b> . Direct sow a few seeds per tube and thin, or sow thinly in trays.                                                                                                                                                                                                                               |
| After sowing          | Water in; keep moist, not wet. Sow promptly after hot-water treatment.                                                                                                                                                                                                                                                                   |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Germination time        | Can start about <b>one week</b> after treatment.                                                            |
| Pricking out            | At the cotyledon to first true leaf stage, before the taproot lengthens.                                    |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem                | Likely cause and fix                                                                        |
|------------------------|---------------------------------------------------------------------------------------------|
| Low germination        | Seed coat not broken. Re-treat unswollen seed or switch between heat and hot-water methods. |
| Seed rots after sowing | Too wet. Water less and improve drainage.                                                   |
| Seedlings collapse     | Damping off. Sterile mix, airflow, avoid crowding.                                          |

### Notes

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| Other notes | Spindly habit – tip-pruning young plants can give a fuller tube.                            |
| Confidence  | <b>High.</b> Nursery germination data and WA hard-seed guidance available for this species. |
| Sources     | See References: 1, 3, 5, 14, 15, 16                                                         |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Grevillea vestita

No widely accepted common name

Family  
Proteaceae

| DIFFICULTY | PRE-TREATMENT               | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|------------|-----------------------------|-------------|--------------|---------------------|
| Moderate   | Nick seed coat; smoke water | Mar – May   | 3–5 mm       | Not recorded        |

|                                    | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|------------------------------------|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| Sowing                             | –   | –   | Best | Best | Best | OK  | OK  | OK  | –   | –   | –   | –   |
| Best window                        |     |     |      |      |      |     |     |     |     |     |     |     |
| Workable – slower or less reliable |     |     |      |      |      |     |     |     |     |     |     |     |
| Avoid / low success                |     |     |      |      |      |     |     |     |     |     |     |     |

## Plant snapshot

|            |                                                                                        |
|------------|----------------------------------------------------------------------------------------|
| Habit      | Erect, spreading, prickly shrub, usually 0.5–3 m.                                      |
| Flowering  | Mainly Jun – Nov (white to cream, often pink-tinged, scented).                         |
| Habitat    | Sandy soils over limestone or granite on the Swan Coastal Plain.                       |
| Local note | Two subspecies exist – label seed by subspecies and provenance. Prickly – wear gloves. |

## Propagation options

| Method   | Suitability | Notes                                                              | Details in       |
|----------|-------------|--------------------------------------------------------------------|------------------|
| Cuttings | Main method | Firm current season's growth strikes reliably.                     | → Cuttings Guide |
| Seed     | Good        | Grows reasonably easily from seed; seedlings vary from the parent. | This guide       |
| Division | Possible    | Can sucker – lift rooted suckers.                                  | –                |

## Seed collection & storage

| Item                  | Details                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Follicles split and drop seed soon after ripening. Tie fine mesh or stocking bags over maturing follicles and collect once they open. |
| Extraction & cleaning | Seed usually falls into the bag; remove follicle debris.                                                                              |
| Seed quality          | Cut-test a few seeds – good seed is firm and white inside.                                                                            |
| Storage life          | Not well documented. Store dry, sealed and cool.                                                                                      |
| Typical germination   | Not recorded. Untreated seed can take 1–2 years to germinate.                                                                         |

## Sowing & growing on

| Item                  | Details                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – May (cool autumn). Jun – Aug workable. Avoid Sep – Feb.                                                                                                                |
| Seed mix              | Free-draining, sterile, low-phosphorus mix.                                                                                                                                  |
| Pre-treatment         | Nick the seed coat with a scalpel (or peel it off after a short soak), then soak in smoke water for 24 hours. Gibberellic acid (GA) can also help (WA Grevillea seed notes). |
| Sowing depth & method | Cover about 3–5 mm. Direct sow into tubes – seedlings have a taproot.                                                                                                        |
| After sowing          | Keep moist, not wet; strict hygiene.                                                                                                                                         |
| Germination time      | A few weeks for treated seed; untreated seed can take 1–2 years.                                                                                                             |
| Pricking out          | If tray-sown, prick out early at the cotyledon stage.                                                                                                                        |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem                      | Likely cause and fix                                               |
|------------------------------|--------------------------------------------------------------------|
| No germination after months  | Seed coat not treated. Nick or peel, then smoke water.             |
| Seedlings yellow or die      | Phosphorus toxicity or root rot. Low-P mix and feed; good hygiene. |
| Seedlings differ from parent | Normal for seed. Use cuttings to keep a particular form.           |

### Notes

|             |                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| Other notes | Cuttings are the main method and keep the parent's form (see the Cuttings Guide).                                 |
| Confidence  | <b>Medium.</b> ANPSA says it grows reasonably easily from seed; treatment follows WA <i>Grevillea</i> seed notes. |
| Sources     | See References: 4, 34, 35                                                                                         |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Hovea pungens

Devil's Pins

Family  
Fabaceae

| DIFFICULTY    | PRE-TREATMENT      | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|---------------|--------------------|-------------|--------------|---------------------|
| Easy-Moderate | Boiling water 30 s | Mar – May   | 4–6 mm       | 41%                 |

|        | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep  | Oct | Nov | Dec |
|--------|-----|-----|------|------|------|-----|-----|-----|------|-----|-----|-----|
| Sowing | –   | OK  | Best | Best | Best | OK  | OK  | OK  | Best | OK  | –   | –   |

|                                                                                               |                                                                                                                      |                                                                                                         |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|  Best window |  Workable – slower or less reliable |  Avoid / low success |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

## Plant snapshot

|            |                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| Habit      | Erect, prickly, broom-like shrub, 0.2–1.8 m.                                                                                  |
| Flowering  | May – Nov (blue-purple).                                                                                                      |
| Habitat    | Wide range: shallow soils on rock, sand, laterite, gravel, clay and loam; granite outcrops, coastal limestone and sandplains. |
| Local note | Recorded in Joondalup and Wanneroo. Sharp leaves – wear gloves when collecting.                                               |

## Propagation options

| Method   | Suitability | Notes               | Details in |
|----------|-------------|---------------------|------------|
| Cuttings | Not used    | Not documented.     | –          |
| Seed     | Main method | Boiling water 30 s. | This guide |
| Division | Not used    | –                   | –          |

## Seed collection & storage

| Item                  | Details                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods ripen in late spring to summer and <b>eject seed when ripe</b> . Bag developing pods with stockings or muslin.                                 |
| Extraction & cleaning | Collect seed from the bags; sieve out pod fragments.                                                                                                |
| Seed quality          | Hard-seeded – fresh seed is dormant until the coat is treated. Float test to remove damaged seed.                                                   |
| Storage life          | <b>Long</b> . Keeps for many years when dry (typical of hard-coated pea-family seed).                                                               |
| Typical germination   | About <b>41%</b> after boiling-water treatment (Alcoa nursery average). About <b>33 seedlings per gram</b> – roughly 3 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> and <b>Sep</b> . Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.                                                                                                                              |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                                                |
| Pre-treatment         | <b>Boiling water:</b> drop into boiling water for <b>30 seconds</b> (Alcoa method), then cool; or pour near-boiling water over the seed and soak overnight. Sow swollen seed; re-treat any that haven't swollen. |
| Sowing depth & method | Larger seed – cover about <b>4–6 mm</b> (roughly twice its thickness). Direct sow 2–3 seeds per tube and thin.                                                                                                   |
| After sowing          | Water in; keep moist, not wet.                                                                                                                                                                                   |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Germination time        | Can start about <b>one week</b> after treatment.                                                            |
| Pricking out            | If tray-sown, prick out at the cotyledon stage – early, as the taproot grows fast.                          |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem                        | Likely cause and fix                                  |
|--------------------------------|-------------------------------------------------------|
| Low germination                | Seed coat not broken. Re-treat unswollen seed.        |
| Swollen seed rots              | Over-boiled or mix too wet. Keep to 30 s; water less. |
| Taproot damage at pricking out | Pricked out too late. Direct sow into tubes instead.  |

### Notes

|             |                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------|
| Other notes | Wear gloves – the pungent (sharp) leaves make handling seedlings and stock plants prickly.              |
| Confidence  | <b>High.</b> Nursery germination data available for this species; method follows WA hard-seed guidance. |
| Sources     | See References: 1, 3, 5, 14, 15, 16                                                                     |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Kennedia prostrata

Scarlet Runner / Running Postman

Family  
Fabaceae

| DIFFICULTY                                                                                                                              | PRE-TREATMENT           |     |      |      | BEST SOWING |     | SOWING DEPTH |     |      | TYPICAL GERMINATION |     |     |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----|------|------|-------------|-----|--------------|-----|------|---------------------|-----|-----|
| Easy                                                                                                                                    | Hot water + smoke water |     |      |      | Mar – May   |     | About 5 mm   |     |      | 30%                 |     |     |
|                                                                                                                                         | Jan                     | Feb | Mar  | Apr  | May         | Jun | Jul          | Aug | Sep  | Oct                 | Nov | Dec |
| Sowing                                                                                                                                  | –                       | OK  | Best | Best | Best        | OK  | OK           | OK  | Best | OK                  | –   | –   |
| <div><div></div> Best window</div> <div><div></div> Workable – slower or less reliable</div> <div><div></div> Avoid / low success</div> |                         |     |      |      |             |     |              |     |      |                     |     |     |

Plant snapshot

|            |                                                             |
|------------|-------------------------------------------------------------|
| Habit      | Prostrate or twining groundcover, spreading to about 2.5 m. |
| Flowering  | Apr – Nov (red pea flowers).                                |
| Habitat    | Usually sandy, gravelly soils.                              |
| Local note | Recorded in Joondalup and Wanneroo.                         |

Propagation options

| Method   | Suitability | Notes                                      | Details in       |
|----------|-------------|--------------------------------------------|------------------|
| Cuttings | Good        | Firm current season's growth strikes well. | → Cuttings Guide |
| Seed     | Main method | Hot water + smoke water; about 8 days.     | This guide       |
| Division | Not used    | –                                          | –                |

Seed collection & storage

| Item                  | Details                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pods (about 50 mm) ripen in late spring to summer and split when ripe – bag them or collect as they turn brown.    |
| Extraction & cleaning | Dry pods in paper bags and collect the seed.                                                                       |
| Seed quality          | Impervious seed coat – dormant until treated. Float test to remove damaged seed.                                   |
| Storage life          | Long. Seed maintains viability for extended periods.                                                               |
| Typical germination   | About 30% after dry heat (100°C, 3 h; Alcoa). About 12 seedlings per gram – roughly 8 g of seed per 100 seedlings. |

Sowing & growing on

| Item                  | Details                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – May and Sep. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.                                                                                                                                                                           |
| Seed mix              | Standard sterile native seed mix.                                                                                                                                                                                                              |
| Pre-treatment         | Hot water: simmer water at 95°C for 2 minutes, pour over the seed (at least 10:1 water to seed) and soak 24 hours, then soak 24 hours in smoke water (Geraldton method). Alternatives: boiling water, abrasion, or dry heat 100°C for 3 hours. |
| Sowing depth & method | Larger seed – cover about 5 mm. Direct sow 1–2 seeds per tube.                                                                                                                                                                                 |
| After sowing          | Water in; keep moist, not wet.                                                                                                                                                                                                                 |
| Germination time      | About 8 days (Geraldton nursery).                                                                                                                                                                                                              |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | If tray-sown, prick out at the cotyledon stage.                                                             |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem                       | Likely cause and fix                           |
|-------------------------------|------------------------------------------------|
| Low germination               | Seed coat not broken. Re-treat unswollen seed. |
| Runners tangle in the nursery | Space tubes out; trim runners before dispatch. |
| Seedlings collapse            | Damping off. Sterile mix, airflow.             |

### Notes

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Other notes | Cuttings of firm current season's growth also strike well.                    |
| Confidence  | <b>High.</b> Nursery data, WA community-nursery method and ANPSA notes agree. |
| Sources     | See References: 3, 4, 5, 14, 24                                               |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
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|           |                       |           |          |        |         |       |

# Leucophyta brownii

Cushion Bush

Family  
Asteraceae

|               |               |             |                       |                     |
|---------------|---------------|-------------|-----------------------|---------------------|
| DIFFICULTY    | PRE-TREATMENT | BEST SOWING | SOWING DEPTH          | TYPICAL GERMINATION |
| Easy-Moderate | None          | Mar – May   | Surface – light cover | Not recorded        |

|                        |     |                                               |      |      |      |     |     |     |     |                                |     |     |
|------------------------|-----|-----------------------------------------------|------|------|------|-----|-----|-----|-----|--------------------------------|-----|-----|
|                        | Jan | Feb                                           | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct                            | Nov | Dec |
| Sowing                 | –   | –                                             | Best | Best | Best | OK  | OK  | OK  | OK  | –                              | –   | –   |
| <div>Best window</div> |     | <div>Workable – slower or less reliable</div> |      |      |      |     |     |     |     | <div>Avoid / low success</div> |     |     |

Plant snapshot

|            |                                                                                             |
|------------|---------------------------------------------------------------------------------------------|
| Habit      | Compact, rounded shrub to about 1 m, densely covered in silvery-white woolly hairs.         |
| Flowering  | Dec or Jan – Feb (small yellow button heads).                                               |
| Habitat    | Coastal foredunes and cliffs.                                                               |
| Local note | Recorded in the Wanneroo LGA. Grows poorly in high humidity and dies easily if waterlogged. |

Propagation options

| Method   | Suitability | Notes                                            | Details in       |
|----------|-------------|--------------------------------------------------|------------------|
| Cuttings | Main method | Semi-hardened cuttings are the preferred method. | → Cuttings Guide |
| Seed     | Good        | No pre-treatment; 10–30 days.                    | This guide       |
| Division | Not used    | –                                                | –                |

Seed collection & storage

| Item                  | Details                                                                       |
|-----------------------|-------------------------------------------------------------------------------|
| Seed collection       | Collect the dry flower heads in late summer, once they turn brown and papery. |
| Extraction & cleaning | Rub heads gently over a sieve to free the small seeds.                        |
| Seed quality          | Many florets may be empty – sow generously.                                   |
| Storage life          | Not well documented. Sow in the first autumn after collection.                |
| Typical germination   | Not recorded. ANBG notes seed germinates 10–30 days after sowing.             |

Sowing & growing on

| Item                    | Details                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Best sowing time        | Mar – May. Jun – Sep workable. Avoid Oct – Feb.                                                      |
| Seed mix                | Standard sterile native seed mix, extra free-draining.                                               |
| Pre-treatment           | None needed (ANBG).                                                                                  |
| Sowing depth & method   | Surface-sow and cover very lightly with fine sand – daisy-family seed often needs light.             |
| After sowing            | Water in with a fine spray; keep just moist. Good airflow.                                           |
| Germination time        | About 10–30 days.                                                                                    |
| Pricking out            | When the second pair of leaves appears.                                                              |
| Shade house & hardening | About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |

| Item                  | Details                                        |
|-----------------------|------------------------------------------------|
| Time to planting size | Est. 3–4 months from sowing (record your own). |

### Common problems

| Problem              | Likely cause and fix                                            |
|----------------------|-----------------------------------------------------------------|
| Seedlings collapse   | Damping off or too humid. Sow thinly, water less, more airflow. |
| Seed buried too deep | Needs light. Surface-sow.                                       |
| Plants die in pots   | Waterlogging. Very free-draining mix; don't overwater.          |

### Notes

|             |                                                                                              |
|-------------|----------------------------------------------------------------------------------------------|
| Other notes | Semi-hardened cuttings are the preferred method (see the Cuttings Guide).                    |
| Confidence  | <b>Medium.</b> ANBG growing notes give treatment and germination time; no nursery test data. |
| Sources     | See References: 4, 30, 31                                                                    |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
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|           |                       |           |          |        |         |       |

# Macrozamia fraseri

Sandplain Zamia

Family  
Zamiaceae (cycads)

| DIFFICULTY  | PRE-TREATMENT                      |      |      |     | BEST SOWING            |     |     | SOWING DEPTH |                     |     | TYPICAL GERMINATION               |     |  |
|-------------|------------------------------------|------|------|-----|------------------------|-----|-----|--------------|---------------------|-----|-----------------------------------|-----|--|
| Moderate    | Remove fleshy coat;<br>float test  |      |      |     | Feb – Mar (fresh seed) |     |     | Surface      |                     |     | ~34% (related <i>M. riedlei</i> ) |     |  |
|             | Jan                                | Feb  | Mar  | Apr | May                    | Jun | Jul | Aug          | Sep                 | Oct | Nov                               | Dec |  |
| Sowing      | OK                                 | Best | Best | OK  | OK                     | –   | –   | –            | –                   | –   | –                                 | –   |  |
| Best window | Workable – slower or less reliable |      |      |     |                        |     |     |              | Avoid / low success |     |                                   |     |  |

## Plant snapshot

|            |                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Habit      | Cycad with a trunk that can exceed 1 m. Male and female cones on separate plants (dioecious).                                                          |
| Flowering  | Cones, not flowers. Seeds with a bright red fleshy coat are shed in late summer to early autumn.                                                       |
| Habitat    | Sand, in heath. Swan Coastal Plain and Geraldton Sandplains.                                                                                           |
| Local note | Recorded in the Wanneroo LGA. <b>All parts, especially the seeds, are poisonous</b> – wear gloves, wash hands, and keep seed away from pets and stock. |

## Propagation options

| Method   | Suitability | Notes                                           | Details in                 |
|----------|-------------|-------------------------------------------------|----------------------------|
| Cuttings | Not used    | Cycad – no cuttings.                            | –                          |
| Seed     | Main method | Sow fresh seed; takes over a year to germinate. | <a href="#">This guide</a> |
| Division | Not used    | Not documented.                                 | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect fallen seed, or seed from cones as they break apart, in <b>late summer to early autumn</b> (February in the 2023 study).                                                                                                                    |
| Extraction & cleaning | Remove the red fleshy coat (sarcotesta). Researchers used a cement mixer with water and 10–30 mm aggregate; small lots can be scrubbed by hand in gloves.                                                                                           |
| Seed quality          | <b>Float test:</b> soak in water for about 5 minutes – seeds that sink are viable; floaters are dead.                                                                                                                                               |
| Storage life          | <b>Short – sow fresh.</b> Seeds are shed moist (about 46% moisture) with an unfinished embryo. Standard cold, dry seed-bank storage (5°C, ~20% RH) damages them: viability was still over 50% at about 6 months but fell to 0–29% after 30+ months. |
| Typical germination   | No figure for this species. The related <i>M. riedlei</i> averaged about <b>34%</b> untreated in Alcoa tests (10 tests). Seeds are large – sow individually.                                                                                        |

## Sowing & growing on

| Item                  | Details                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>As soon as seed is cleaned</b> – usually Feb – Mar. Jan and Apr – May are workable.                                                |
| Seed mix              | Coarse, free-draining, sand-based mix in <b>deep containers</b> – cycads have a long root.                                            |
| Pre-treatment         | None beyond removing the fleshy coat and float testing.                                                                               |
| Sowing depth & method | <b>Surface-sow:</b> lay seeds on the mix and press them in slightly; don't bury. Sow one seed per deep tube or pot, or in deep trays. |

| Item                    | Details                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| After sowing            | Keep outdoors under natural temperatures and rainfall, as the researchers recommend. Water lightly in dry spells – avoid heavy irrigation.                  |
| Germination time        | Slow. The embryo keeps developing after the seed falls; the first root appears after about 5 months and full germination takes <b>more than 12 months</b> . |
| Pricking out            | Avoid if possible – sow direct into deep containers. If tray-sown, pot each seed on once its root appears, keeping the root intact.                         |
| Shade house & hardening | Grow on in light shade and harden off to full sun over several weeks once the first leaf is firm.                                                           |
| Time to planting size   | Very slow – allow 1–2 years or more from sowing (record your own).                                                                                          |

### Common problems

| Problem                        | Likely cause and fix                                                                |
|--------------------------------|-------------------------------------------------------------------------------------|
| Seed rots                      | Too wet, or damaged when the fleshy coat was removed. Water lightly; handle gently. |
| No sign of life after 6 months | Normal – germination takes over a year. Keep trays going.                           |
| Poor results from stored seed  | Seed dried or cold-stored. Sow fresh, within a few months of collection.            |

### Notes

|             |                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Other notes | Poisonous: label trays clearly and keep seed and seedlings out of reach of children, pets and stock.                                              |
| Confidence  | <b>High.</b> Species-specific seed storage and germination research published in 2023. Germination figure is from the related <i>M. riedlei</i> . |
| Sources     | See References: 3, 4, 26, 27                                                                                                                      |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Melaleuca huegelii

Chenille Honeymyrtle

Family  
Myrtaceae

| DIFFICULTY | PRE-TREATMENT | BEST SOWING          | SOWING DEPTH | TYPICAL GERMINATION |
|------------|---------------|----------------------|--------------|---------------------|
| Easy       | None          | Mar – Apr, Sep – Oct | Surface      | 58%                 |

|        | Jan         | Feb                                | Mar  | Apr  | May | Jun | Jul | Aug | Sep  | Oct  | Nov | Dec                 |
|--------|-------------|------------------------------------|------|------|-----|-----|-----|-----|------|------|-----|---------------------|
| Sowing | –           | OK                                 | Best | Best | OK  | OK  | OK  | OK  | Best | Best | OK  | –                   |
|        | Best window | Workable – slower or less reliable |      |      |     |     |     |     |      |      |     | Avoid / low success |

Plant snapshot

|            |                                                   |
|------------|---------------------------------------------------|
| Habit      | Shrub or small tree, 0.5–5 m.                     |
| Flowering  | Sep – Dec or Jan (pink, white or pink-purple).    |
| Habitat    | Sand; limestone cliffs, coastal plains and dunes. |
| Local note | Recorded in Joondalup and Wanneroo.               |

Propagation options

| Method   | Suitability | Notes                                        | Details in       |
|----------|-------------|----------------------------------------------|------------------|
| Cuttings | Good        | Reliable in this genus; keeps a chosen form. | → Cuttings Guide |
| Seed     | Main method | Surface-sow fine seed; no pre-treatment.     | This guide       |
| Division | Not used    | Woody shrub.                                 | –                |

Seed collection & storage

| Item                  | Details                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Seed stays in the woody capsules indefinitely, so capsules can be collected at any time. Use capsules at least 12 months old. |
| Extraction & cleaning | Place capsules in an open container somewhere warm; fine seed is released in 2–3 days.                                        |
| Seed quality          | Extremely fine seed with chaff; no need to separate.                                                                          |
| Storage life          | Long if kept dry and cool; seed also keeps on the plant in closed capsules.                                                   |
| Typical germination   | About 58% (untreated, Alcoa nursery average). About 775 seedlings per gram – a pinch is plenty.                               |

Sowing & growing on

| Item                  | Details                                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan.                                                  |
| Seed mix              | Fine-textured, sterile native seed mix.                                                                                     |
| Pre-treatment         | None needed.                                                                                                                |
| Sowing depth & method | Mix with dry, sterile yellow sand and sprinkle from a spice jar. Sow on the surface; don't cover.                           |
| After sowing          | Water in with smoke water using a fine mist (Geraldton), or stand pots in a saucer of water until germination (bog method). |
| Germination time      | About 9 days at Geraldton; 14–30 days is typical for the genus.                                                             |
| Pricking out          | When the second pair of leaves appears. Handle tiny seedlings by the leaves.                                                |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem               | Likely cause and fix                        |
|-----------------------|---------------------------------------------|
| Seedlings too crowded | Dilute seed more with sand.                 |
| Seed washed away      | Use bottom watering or a very fine mist.    |
| Damping off           | Crowding and wet mix. Thin sowing, airflow. |

### Notes

|             |                                                                                     |
|-------------|-------------------------------------------------------------------------------------|
| Other notes | Cuttings are also reliable if a particular form is wanted.                          |
| Confidence  | <b>High.</b> Nursery data, WA community-nursery method and ANPSA genus notes agree. |
| Sources     | See References: 3, 4, 5, 18                                                         |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Melaleuca systema

Coastal Honeymyrtle

Family  
Myrtaceae

|               |               |                      |              |                     |
|---------------|---------------|----------------------|--------------|---------------------|
| DIFFICULTY    | PRE-TREATMENT | BEST SOWING          | SOWING DEPTH | TYPICAL GERMINATION |
| Easy-Moderate | None          | Mar – Apr, Sep – Oct | Surface      | 14%                 |

|                        |     |                                               |      |      |     |     |     |     |      |      |     |                                |
|------------------------|-----|-----------------------------------------------|------|------|-----|-----|-----|-----|------|------|-----|--------------------------------|
|                        | Jan | Feb                                           | Mar  | Apr  | May | Jun | Jul | Aug | Sep  | Oct  | Nov | Dec                            |
| Sowing                 | –   | OK                                            | Best | Best | OK  | OK  | OK  | OK  | Best | Best | OK  | –                              |
| <div>Best window</div> |     | <div>Workable – slower or less reliable</div> |      |      |     |     |     |     |      |      |     | <div>Avoid / low success</div> |

Plant snapshot

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| Habit      | Erect to spreading shrub, usually 0.5–2 m.                                     |
| Flowering  | Feb – Mar or Aug – Dec (yellow-cream).                                         |
| Habitat    | Sand over laterite or limestone; stabilised coastal dunes and rocky limestone. |
| Local note | Recorded in Joondalup and Wanneroo.                                            |

Propagation options

|          |             |                                                                   |                  |
|----------|-------------|-------------------------------------------------------------------|------------------|
| Method   | Suitability | Notes                                                             | Details in       |
| Cuttings | Good        | Reliable in this genus; a good option given low seed germination. | → Cuttings Guide |
| Seed     | Main method | Surface-sow fine seed; no pre-treatment.                          | This guide       |
| Division | Not used    | Woody shrub.                                                      | –                |

Seed collection & storage

|                       |                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Item                  | Details                                                                                                    |
| Seed collection       | Collect capsules at least 12 months old from older stems at any time; seed is retained in closed capsules. |
| Extraction & cleaning | Place capsules somewhere warm in an open container; seed is released in 2–3 days.                          |
| Seed quality          | Fine seed with chaff. Germination in nursery tests was low – sow generously.                               |
| Storage life          | Long if kept dry and cool.                                                                                 |
| Typical germination   | About 14% (untreated, Alcoa average of only 3 tests). About 301 seedlings per gram.                        |

Sowing & growing on

|                       |                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------|
| Item                  | Details                                                                           |
| Best sowing time      | Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan.        |
| Seed mix              | Fine-textured, sterile native seed mix.                                           |
| Pre-treatment         | None needed.                                                                      |
| Sowing depth & method | Mix with dry sand and sprinkle from a spice jar. Sow on the surface; don't cover. |
| After sowing          | Water in with smoke water using a fine mist, or use the bog method.               |
| Germination time      | About 7 days at Geraldton; allow up to 4 weeks.                                   |
| Pricking out          | When the second pair of leaves appears.                                           |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem          | Likely cause and fix                                                               |
|------------------|------------------------------------------------------------------------------------|
| Low germination  | Empty or old seed. Use older capsules (12+ months) from vigorous plants; sow more. |
| Seed washed away | Use bottom watering or a very fine mist.                                           |
| Damping off      | Crowding and wet mix.                                                              |

### Notes

|             |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------|
| Other notes | Cuttings are a reliable alternative for this genus.                                                |
| Confidence  | <b>Medium.</b> Few nursery tests (3); WA community-nursery method and ANPSA genus notes available. |
| Sources     | See References: 3, 4, 5, 18                                                                        |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Myoporum insulare

Blueberry Tree / Common Boobialla

Family  
Scrophulariaceae

| DIFFICULTY | PRE-TREATMENT                        | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|------------|--------------------------------------|-------------|--------------|---------------------|
| Moderate   | Ferment berries;<br>smoke-water soak | Mar – May   | 2–3 mm       | Not recorded        |

|                                    | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|------------------------------------|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| Sowing                             | –   | –   | Best | Best | Best | OK  | OK  | OK  | OK  | –   | –   | –   |
| Best window                        |     |     |      |      |      |     |     |     |     |     |     |     |
| Workable – slower or less reliable |     |     |      |      |      |     |     |     |     |     |     |     |
| Avoid / low success                |     |     |      |      |      |     |     |     |     |     |     |     |

### Plant snapshot

|            |                                                                                              |
|------------|----------------------------------------------------------------------------------------------|
| Habit      | Dense, low spreading to erect shrub or small tree, 0.25–5 m, with thick, fleshy leaves.      |
| Flowering  | Jul – Dec, also Jan – Feb (white, purple-spotted), followed by purple berries.               |
| Habitat    | White or grey sand on coastal dunes and limestone.                                           |
| Local note | Recorded in Joondalup and Wanneroo. Forms range from prostrate to tree-like; seedlings vary. |

### Propagation options

| Method   | Suitability | Notes                                                           | Details in       |
|----------|-------------|-----------------------------------------------------------------|------------------|
| Cuttings | Main method | Firm young growth roots readily.                                | → Cuttings Guide |
| Seed     | Good        | Ferment berries, smoke water; 15–30 days. Seedling form varies. | This guide       |
| Division | Not used    | Woody shrub.                                                    | –                |

### Seed collection & storage

| Item                  | Details                                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pick the <b>darkest purple</b> , fully ripe berries.                                                               |
| Extraction & cleaning | <b>Ferment:</b> soak the berries in water for 4–5 days, then wash off the pulp through a sieve and dry the stones. |
| Seed quality          | Hard stones; cut-test a few to check they are filled.                                                              |
| Storage life          | Not well documented. Store dry and cool.                                                                           |
| Typical germination   | Not recorded.                                                                                                      |

### Sowing & growing on

| Item                  | Details                                                                             |
|-----------------------|-------------------------------------------------------------------------------------|
| Best sowing time      | Mar – May. Jun – Sep workable. Avoid Oct – Feb.                                     |
| Seed mix              | Standard sterile native seed mix.                                                   |
| Pre-treatment         | After fermenting, <b>soak in smoke water (1:9) for 24 hours</b> (Geraldton method). |
| Sowing depth & method | Cover about <b>2–3 mm</b> . Sow in trays or 2–3 stones per tube.                    |
| After sowing          | Water in; keep moist, not wet.                                                      |
| Germination time      | About <b>15–30 days</b> (Geraldton nursery).                                        |
| Pricking out          | When the second pair of leaves appears.                                             |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–5 months from sowing (record your own).                                                              |

### Common problems

| Problem                        | Likely cause and fix                                                       |
|--------------------------------|----------------------------------------------------------------------------|
| Poor germination               | Unripe berries or pulp left on. Use the darkest berries and ferment fully. |
| Mould on trays                 | Pulp residue. Wash stones well before sowing.                              |
| Plants not the form you wanted | Seedlings vary. Use cuttings for a particular form.                        |

### Notes

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Other notes | Cuttings are the main method and keep the parent's form (see the Cuttings Guide). |
| Confidence  | <b>Medium.</b> WA community-nursery method available; no nursery test data.       |
| Sources     | See References: 4, 5                                                              |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Olearia axillaris

Coastal Daisybush

Family  
Asteraceae

| DIFFICULTY    | PRE-TREATMENT                    | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|---------------|----------------------------------|-------------|--------------|---------------------|
| Easy-Moderate | Air-dry 3 months;<br>smoke water | Aug – Sep   | Surface      | 21%                 |

|        | Jan         | Feb | Mar                                | Apr | May | Jun | Jul | Aug  | Sep  | Oct | Nov | Dec                 |
|--------|-------------|-----|------------------------------------|-----|-----|-----|-----|------|------|-----|-----|---------------------|
| Sowing | –           | –   | OK                                 | OK  | OK  | OK  | OK  | Best | Best | OK  | –   | –                   |
|        | Best window |     | Workable – slower or less reliable |     |     |     |     |      |      |     |     | Avoid / low success |

Plant snapshot

|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| Habit      | Erect, much-branched grey shrub, 0.2–3 m.                                            |
| Flowering  | Jan – Jul or Nov – Dec (small white to yellow daisies).                              |
| Habitat    | White, grey or red sand and loam; coastal limestone, sand dunes and rocky hillsides. |
| Local note | Recorded in Joondalup and Wanneroo.                                                  |

Propagation options

| Method   | Suitability | Notes                                                  | Details in       |
|----------|-------------|--------------------------------------------------------|------------------|
| Cuttings | Good        | Perennial daisies strike quickly from vigorous growth. | → Cuttings Guide |
| Seed     | Main method | Fresh seed, aired 3 months; surface-sow.               | This guide       |
| Division | Not used    | Woody shrub.                                           | –                |

Seed collection & storage

| Item                  | Details                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Seed usually ripens around <b>April – May</b> , but plants can set seed about <b>6 weeks after a big rain</b> , so watch for opportunistic crops (e.g. after summer storms). |
| Extraction & cleaning | Collect the fluffy seed heads into paper bags.                                                                                                                               |
| Seed quality          | Germination in nursery tests was moderate (about 21%).                                                                                                                       |
| Storage life          | <b>Short – needs fresh seed each year.</b> Seed must be <b>aired for about 3 months</b> before storing or sowing.                                                            |
| Typical germination   | About <b>21%</b> (untreated, Alcoa nursery average). About <b>73 seedlings per gram</b> – roughly 1.4 g of seed per 100 seedlings.                                           |

Sowing & growing on

| Item                  | Details                                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Aug – Sep</b> – about 3 months after an autumn collection. Mar – Jul and Oct workable depending on when seed was collected. Avoid Nov – Feb. |
| Seed mix              | Standard sterile native seed mix.                                                                                                               |
| Pre-treatment         | Air the seed for about 3 months after collection. Water trays with smoke water after sowing (Geraldton method).                                 |
| Sowing depth & method | <b>Sow on the surface</b> and press in lightly – daisy-family seed often needs light to germinate.                                              |
| After sowing          | Water in with smoke water; keep moist.                                                                                                          |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Germination time        | About <b>7–16 days</b> (Geraldton nursery).                                                                 |
| Pricking out            | When the second pair of leaves appears.                                                                     |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem              | Likely cause and fix                                               |
|----------------------|--------------------------------------------------------------------|
| Poor germination     | Old seed or not aired. Use this season's seed, aired for 3 months. |
| Seed buried too deep | Needs light. Surface sow.                                          |
| Leggy seedlings      | Too much shade. Move to full sun sooner.                           |

### Notes

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| Other notes | Also strikes readily from cuttings.                                             |
| Confidence  | <b>High.</b> Nursery data and a detailed WA community-nursery method available. |
| Sources     | See References: 3, 4, 5, 23                                                     |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
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|           |                       |           |          |        |         |       |

# Petrophile linearis

Pixie Mops

Family  
Proteaceae

| DIFFICULTY                                                                                                   | PRE-TREATMENT                  |     | BEST SOWING |      |      | SOWING DEPTH |     |     | TYPICAL GERMINATION |     |     |     |
|--------------------------------------------------------------------------------------------------------------|--------------------------------|-----|-------------|------|------|--------------|-----|-----|---------------------|-----|-----|-----|
| Difficult                                                                                                    | Remove hairs, nick coat; smoke |     | Mar – May   |      |      | 3–5 mm       |     |     | <1% untreated       |     |     |     |
|                                                                                                              | Jan                            | Feb | Mar         | Apr  | May  | Jun          | Jul | Aug | Sep                 | Oct | Nov | Dec |
| Sowing                                                                                                       | –                              | –   | Best        | Best | Best | OK           | OK  | OK  | –                   | –   | –   | –   |
| <div><div>Best window</div><div>Workable – slower or less reliable</div><div>Avoid / low success</div></div> |                                |     |             |      |      |              |     |     |                     |     |     |     |

Plant snapshot

|            |                                                              |
|------------|--------------------------------------------------------------|
| Habit      | Erect shrub, 0.2–1 m.                                        |
| Flowering  | Aug – Dec (pink and white ‘mops’).                           |
| Habitat    | White, grey or yellow sand, often over laterite; sandplains. |
| Local note | Recorded in Joondalup and Wanneroo.                          |

Propagation options

| Method   | Suitability | Notes                     | Details in |
|----------|-------------|---------------------------|------------|
| Cuttings | Not used    | Not documented.           | –          |
| Seed     | Difficult   | Nick the seed coat; slow. | This guide |
| Division | Not used    | –                         | –          |

Seed collection & storage

| Item                  | Details                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Cones are usually ripe about <b>six months after flowering</b> , when they turn grey to grey-brown and lose their old floral parts. |
| Extraction & cleaning | Seed does not come out easily in <i>Petrophile</i> – you may need to pull the hairy nuts out of the cone with forceps.              |
| Seed quality          | Very low germination untreated (0.17% in Alcoa tests). Many nuts may be empty – cut-test.                                           |
| Storage life          | Not well documented. Store dry and cool.                                                                                            |
| Typical germination   | Untreated: <b>under 1%</b> (Alcoa, 6 tests). Treated seed should do much better – record your results.                              |

Sowing & growing on

| Item                  | Details                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> (cool autumn). Jun – Aug workable. Avoid Sep – Feb.                                                                                     |
| Seed mix              | Free-draining, sterile, <b>low-phosphorus</b> mix.                                                                                                       |
| Pre-treatment         | <b>Remove the hairs and nick a small section of the seed coat</b> to reveal the white seed (without damaging it). Smoke water (24 h soak) may also help. |
| Sowing depth & method | Cover about <b>3–5 mm</b> . Direct sow into tubes to avoid root damage.                                                                                  |
| After sowing          | Keep moist, not wet; strict hygiene.                                                                                                                     |
| Germination time      | Can take <b>several months</b> – keep trays going.                                                                                                       |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | If tray-sown, prick out at the cotyledon stage.                                                             |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 6–8 months from sowing (record your own).                                                              |

### Common problems

| Problem                 | Likely cause and fix                                           |
|-------------------------|----------------------------------------------------------------|
| No germination          | Seed coat not nicked, or empty seed. Nick carefully; cut-test. |
| Seed damaged by nicking | Cut too deep. Remove just enough coat to see white seed.       |
| Seedlings die in pots   | Phosphorus toxicity or root rot. Low-P mix and fertiliser.     |

### Notes

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| Other notes | A slow, fiddly species from seed – set larger batches and keep good records.                               |
| Confidence  | <b>Medium.</b> Nursery tests show very low untreated germination; method follows WA Petrophile seed notes. |
| Sources     | See References: 3, 4, 21                                                                                   |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Petrophile macrostachya

No widely used common name

Family  
Proteaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT                         |     |      |      | BEST SOWING      |     |     | SOWING DEPTH  |     | TYPICAL GERMINATION |     |     |  |
|-----------------------------------------------------------------------------------------------------|---------------------------------------|-----|------|------|------------------|-----|-----|---------------|-----|---------------------|-----|-----|--|
| <b>Difficult *</b>                                                                                  | <b>Remove hairs, nick coat; smoke</b> |     |      |      | <b>Mar – May</b> |     |     | <b>3–5 mm</b> |     | <b>Not recorded</b> |     |     |  |
|                                                                                                     | Jan                                   | Feb | Mar  | Apr  | May              | Jun | Jul | Aug           | Sep | Oct                 | Nov | Dec |  |
| <b>Sowing</b>                                                                                       | –                                     | –   | Best | Best | Best             | OK  | OK  | OK            | –   | –                   | –   | –   |  |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                                       |     |      |      |                  |     |     |               |     |                     |     |     |  |

## Plant snapshot

|                   |                                                        |
|-------------------|--------------------------------------------------------|
| <b>Habit</b>      | Erect, compact or spreading, prickly shrub, 0.3–1.2 m. |
| <b>Flowering</b>  | Jul – Nov (yellow-cream).                              |
| <b>Habitat</b>    | White or yellow-brown sand and gravel; sandplains.     |
| <b>Local note</b> | Recorded in the Wanneroo LGA. Prickly – wear gloves.   |

## Propagation options

| Method   | Suitability      | Notes                             | Details in        |
|----------|------------------|-----------------------------------|-------------------|
| Cuttings | Not used         | Not documented.                   | –                 |
| Seed     | <b>Difficult</b> | Nick the seed coat; slow (trial). | <b>This guide</b> |
| Division | Not used         | –                                 | –                 |

## Seed collection & storage

| Item                             | Details                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Seed collection</b>           | Cones are usually ripe about six months after flowering, when grey to grey-brown and the old floral parts have gone. |
| <b>Extraction &amp; cleaning</b> | Pull the hairy nuts from the cones with forceps.                                                                     |
| <b>Seed quality</b>              | Cut-test a sample – expect many empty nuts.                                                                          |
| <b>Storage life</b>              | Not well documented. Store dry and cool.                                                                             |
| <b>Typical germination</b>       | Not recorded. Expect low germination unless the seed coat is treated.                                                |

## Sowing & growing on

| Item                             | Details                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------|
| <b>Best sowing time</b>          | <b>Mar – May.</b> Jun – Aug workable. Avoid Sep – Feb.                                 |
| <b>Seed mix</b>                  | Free-draining, sterile, low-phosphorus mix.                                            |
| <b>Pre-treatment</b>             | Remove the hairs and nick a small section of the seed coat; smoke water may also help. |
| <b>Sowing depth &amp; method</b> | Cover about <b>3–5 mm</b> . Direct sow into tubes.                                     |
| <b>After sowing</b>              | Keep moist, not wet; strict hygiene.                                                   |
| <b>Germination time</b>          | Can take <b>several months</b> .                                                       |
| <b>Pricking out</b>              | If tray-sown, prick out at the cotyledon stage.                                        |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 6–8 months from sowing (record your own).                                                              |

### Common problems

| Problem                 | Likely cause and fix                 |
|-------------------------|--------------------------------------|
| No germination          | Seed coat not nicked, or empty seed. |
| Seed damaged by nicking | Cut less deeply.                     |
| Seedlings die in pots   | Phosphorus toxicity or root rot.     |

### Notes

|             |                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------|
| Other notes | Treat as a trial – record results so this page can be updated.                                                               |
| Confidence  | <b>Low.</b> No species data found. Method based on WA Petrophile seed notes and <i>P. linearis</i> results – trial protocol. |
| Sources     | See References: 4, 21                                                                                                        |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# *Podotheca gnaphalioides*

Golden Long-heads

Family  
Asteraceae

| DIFFICULTY | PRE-TREATMENT          |     |                                    | BEST SOWING |      |     | SOWING DEPTH          |     |     | TYPICAL GERMINATION |     |                     |
|------------|------------------------|-----|------------------------------------|-------------|------|-----|-----------------------|-----|-----|---------------------|-----|---------------------|
| Moderate   | Smoke water (optional) |     |                                    | Apr – May   |      |     | Surface – light cover |     |     | 7%                  |     |                     |
|            | Jan                    | Feb | Mar                                | Apr         | May  | Jun | Jul                   | Aug | Sep | Oct                 | Nov | Dec                 |
| Sowing     | –                      | –   | OK                                 | Best        | Best | OK  | –                     | –   | –   | –                   | –   | –                   |
|            | Best window            |     | Workable – slower or less reliable |             |      |     |                       |     |     |                     |     | Avoid / low success |

## Plant snapshot

|            |                                                                                                |
|------------|------------------------------------------------------------------------------------------------|
| Habit      | Annual herb, 0.02–0.6 m.                                                                       |
| Flowering  | Aug – Nov (golden-yellow, long narrow heads).                                                  |
| Habitat    | Sandy soils.                                                                                   |
| Local note | Recorded in Joondalup and Wanneroo. An annual – it flowers the spring after sowing, then dies. |

## Propagation options

| Method   | Suitability | Notes                                                  | Details in                 |
|----------|-------------|--------------------------------------------------------|----------------------------|
| Cuttings | Not used    | Annual.                                                | –                          |
| Seed     | Main method | Annual everlasting – sow after the first autumn rains. | <a href="#">This guide</a> |
| Division | Not used    | Annual.                                                | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect heads in late spring–early summer once they dry and the fluffy seed starts to lift.                                         |
| Extraction & cleaning | Rub heads gently over a sieve. The seed can be sown with its fluff.                                                                 |
| Seed quality          | Germination in nursery tests was low (about 7%) – sow generously.                                                                   |
| Storage life          | Store dry and cool and sow the following autumn.                                                                                    |
| Typical germination   | About 7% (untreated, Alcoa nursery average, 8 tests). About <b>84 seedlings per gram</b> – roughly 1.2 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Apr – May</b> , after the first autumn rains. Mar and Jun are workable. Later sowings flower poorly.                                |
| Seed mix              | Standard sterile native seed mix.                                                                                                      |
| Pre-treatment         | None required. Smoke water in the first watering may help (it helps many annual daisies).                                              |
| Sowing depth & method | Surface-sow and cover very lightly – daisy seed often needs light. Sowing a pinch straight into each tube or cell avoids pricking out. |
| After sowing          | Water in with a fine spray; keep moist.                                                                                                |
| Germination time      | About <b>2–4 weeks</b> .                                                                                                               |
| Pricking out          | If tray-sown, prick out small, at the second-leaf stage. Thin direct-sown cells to 1–3 seedlings.                                      |

| Item                    | Details                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | Short – a week or so in the shade house, then full sun.                                                    |
| Time to planting size   | Plant out small, about <b>6–10 weeks</b> after sowing, so plants establish in winter and flower in spring. |

### Common problems

| Problem                   | Likely cause and fix                                   |
|---------------------------|--------------------------------------------------------|
| Very low germination      | Normal for this species. Sow thickly; try smoke water. |
| Leggy seedlings           | Too much shade. Move to full sun early.                |
| Plants flower in the tube | Held too long. Plant out while small.                  |

### Notes

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| Other notes | Often direct-seeded into revegetation sites after the first autumn rains.                                    |
| Confidence  | <b>Medium.</b> Nursery germination data (8 tests); sowing follows standard practice for annual everlastings. |
| Sources     | See References: 3, 4, 30                                                                                     |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Ptilotus drummondii

Narrowleaf Mulla Mulla

Family  
Amaranthaceae

| DIFFICULTY                                                                                                   | PRE-TREATMENT                     |     |      |      | BEST SOWING |     | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |     |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------|-----|------|------|-------------|-----|--------------|-----|---------------------|-----|-----|-----|
| Moderate-Difficult                                                                                           | Remove woolly floral parts; smoke |     |      |      | Mar – May   |     | 1–2 mm       |     | 4%                  |     |     |     |
|                                                                                                              | Jan                               | Feb | Mar  | Apr  | May         | Jun | Jul          | Aug | Sep                 | Oct | Nov | Dec |
| Sowing                                                                                                       | –                                 | OK  | Best | Best | Best        | OK  | OK           | OK  | OK                  | –   | –   | –   |
| <div><div>Best window</div><div>Workable – slower or less reliable</div><div>Avoid / low success</div></div> |                                   |     |      |      |             |     |              |     |                     |     |     |     |

Plant snapshot

|            |                                                              |
|------------|--------------------------------------------------------------|
| Habit      | Perennial herb, 0.05–0.8 m.                                  |
| Flowering  | Mar, May and Jul – Dec (white, cream or pink woolly spikes). |
| Habitat    | Sandy soils.                                                 |
| Local note | Recorded in the Wanneroo LGA.                                |

Propagation options

| Method   | Suitability | Notes                                                               | Details in |
|----------|-------------|---------------------------------------------------------------------|------------|
| Cuttings | Possible    | Cuttings have worked in the genus; not documented for this species. | –          |
| Seed     | Main method | Remove the woolly floral parts to lift germination.                 | This guide |
| Division | Not used    | –                                                                   | –          |

Seed collection & storage

| Item                  | Details                                                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Seed is held inside the woolly flowers. Collect spikes when the flowers are <b>papery and very dry</b> and rub off easily.                                                               |
| Extraction & cleaning | Rub spikes over a sieve to separate the individual flowers – each holds at most one seed.                                                                                                |
| Seed quality          | Many flowers are empty. Check a few by rubbing off the wool.                                                                                                                             |
| Storage life          | Not well documented. Store dry and cool.                                                                                                                                                 |
| Typical germination   | Untreated: about 4% (Alcoa, 29 tests), about 12 seedlings per gram of whole flowers. In the related <i>P. exaltatus</i> , removing the woolly floral parts raised germination to 60–80%. |

Sowing & growing on

| Item                  | Details                                                                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | Mar – May. Feb and Jun – Sep are workable. Avoid Oct – Jan.                                                                                                                                                                                         |
| Seed mix              | Standard sterile seed mix, extra free-draining.                                                                                                                                                                                                     |
| Pre-treatment         | Remove the woolly floral parts by rubbing the flowers gently on a sieve or between rough boards to free the seed. Heat and cold treatments did not help in <i>P. exaltatus</i> . Water in with smoke water (Geraldton method for <i>Ptilotus</i> ). |
| Sowing depth & method | Cover about 1–2 mm. Sow cleaned seed thinly, or 2–3 seeds direct into tubes.                                                                                                                                                                        |
| After sowing          | Keep just moist – <i>Ptilotus</i> rot easily if wet or humid.                                                                                                                                                                                       |
| Germination time      | About 6–10 days for other <i>Ptilotus</i> at Geraldton; untreated seed is slow and erratic.                                                                                                                                                         |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | Early and carefully, at the first true leaves – or sow direct into tubes.                                   |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem                    | Likely cause and fix                                      |
|----------------------------|-----------------------------------------------------------|
| Almost no germination      | Wool left on, or empty flowers. Clean the seed; sow more. |
| Seedlings collapse         | Too wet. Water less; good drainage and airflow.           |
| Plants short-lived in pots | Humidity and wet mix. Full sun, very free-draining mix.   |

### Notes

|             |                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Other notes | Cuttings have been successful in the genus (ANPSA) – worth trialling from non-flowering shoots.                                       |
| Confidence  | <b>Medium.</b> Nursery test data (29 tests) show low untreated germination; treatment is based on research with <i>P. exaltatus</i> . |
| Sources     | See References: 3, 4, 5, 28, 29                                                                                                       |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# *Ptilotus manglesii*

Pom Poms

Family  
Amaranthaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT                     |     |      |      | BEST SOWING |     | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |     |
|-----------------------------------------------------------------------------------------------------|-----------------------------------|-----|------|------|-------------|-----|--------------|-----|---------------------|-----|-----|-----|
| Moderate-Difficult                                                                                  | Remove woolly floral parts; smoke |     |      |      | Mar – May   |     | 1–2 mm       |     | 3%                  |     |     |     |
|                                                                                                     | Jan                               | Feb | Mar  | Apr  | May         | Jun | Jul          | Aug | Sep                 | Oct | Nov | Dec |
| Sowing                                                                                              | –                                 | OK  | Best | Best | Best        | OK  | OK           | OK  | OK                  | –   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                                   |     |      |      |             |     |              |     |                     |     |     |     |

## Plant snapshot

|            |                                                                                  |
|------------|----------------------------------------------------------------------------------|
| Habit      | Low perennial herb, 0.05–0.3 m.                                                  |
| Flowering  | Sep – Dec or Jan (pink to magenta pom-pom heads).                                |
| Habitat    | Sandy and gravelly soils.                                                        |
| Local note | Recorded in Joondalup and Wanneroo. Showy, but often short-lived in cultivation. |

## Propagation options

| Method   | Suitability | Notes                                                               | Details in                 |
|----------|-------------|---------------------------------------------------------------------|----------------------------|
| Cuttings | Possible    | Cuttings have worked in the genus; not documented for this species. | –                          |
| Seed     | Main method | Remove the woolly floral parts to lift germination.                 | <a href="#">This guide</a> |
| Division | Not used    | –                                                                   | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect heads when the flowers are <b>papery and very dry</b> (early summer).                                                                                                                 |
| Extraction & cleaning | Rub heads over a sieve to separate the individual flowers.                                                                                                                                    |
| Seed quality          | Many flowers are empty. Check a few by rubbing off the wool.                                                                                                                                  |
| Storage life          | Not well documented. Store dry and cool.                                                                                                                                                      |
| Typical germination   | Untreated: about <b>3%</b> (Alcoa, 11 tests), about 6 seedlings per gram of whole flowers. Removing the woolly floral parts raised germination to 60–80% in the related <i>P. exaltatus</i> . |

## Sowing & growing on

| Item                  | Details                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May</b> . Feb and Jun – Sep are workable. Avoid Oct – Jan.                                  |
| Seed mix              | Standard sterile seed mix, extra free-draining.                                                      |
| Pre-treatment         | <b>Remove the woolly floral parts</b> to free the seed, then water in with smoke water after sowing. |
| Sowing depth & method | Cover about <b>1–2 mm</b> . Sow cleaned seed thinly, or direct into tubes.                           |
| After sowing          | Keep just moist; good airflow.                                                                       |
| Germination time      | About <b>6–10 days</b> for other <i>Ptilotus</i> at Geraldton; untreated seed is slow and erratic.   |
| Pricking out          | Early and carefully, at the first true leaves – or sow direct into tubes.                            |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem                     | Likely cause and fix                                      |
|-----------------------------|-----------------------------------------------------------|
| Almost no germination       | Wool left on, or empty flowers. Clean the seed; sow more. |
| Seedlings collapse          | Too wet. Water less; improve drainage.                    |
| Plants die after potting on | Overwatering or humidity. Full sun, free-draining mix.    |

### Notes

|             |                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Other notes | Cuttings have been successful in the genus (ANPSA) – worth trialling.                                                                 |
| Confidence  | <b>Medium.</b> Nursery test data (11 tests) show low untreated germination; treatment is based on research with <i>P. exaltatus</i> . |
| Sources     | See References: 3, 4, 5, 28, 29                                                                                                       |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Ptilotus polystachyus

Prince of Wales Feather

Family  
Amaranthaceae

| DIFFICULTY                                                                                                   | PRE-TREATMENT                     |     |      |      | BEST SOWING |     |     | SOWING DEPTH |     | TYPICAL GERMINATION |     |     |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------|-----|------|------|-------------|-----|-----|--------------|-----|---------------------|-----|-----|
| Moderate *                                                                                                   | Remove woolly floral parts; smoke |     |      |      | Mar – May   |     |     | 1–2 mm       |     | Not recorded        |     |     |
|                                                                                                              | Jan                               | Feb | Mar  | Apr  | May         | Jun | Jul | Aug          | Sep | Oct                 | Nov | Dec |
| Sowing                                                                                                       | –                                 | OK  | Best | Best | Best        | OK  | OK  | OK           | OK  | –                   | –   | –   |
| <div><div>Best window</div><div>Workable – slower or less reliable</div><div>Avoid / low success</div></div> |                                   |     |      |      |             |     |     |              |     |                     |     |     |

Plant snapshot

|            |                                                                        |
|------------|------------------------------------------------------------------------|
| Habit      | Annual or short-lived perennial herb, 0.15–1.5 m.                      |
| Flowering  | May – Nov or Jan (green, white or yellow feathery spikes).             |
| Habitat    | Sandy soils.                                                           |
| Local note | Recorded in the Perth area (Queens Park). Short-lived – sow each year. |

Propagation options

| Method   | Suitability | Notes                                                               | Details in |
|----------|-------------|---------------------------------------------------------------------|------------|
| Cuttings | Possible    | Cuttings have worked in the genus; not documented for this species. | –          |
| Seed     | Main method | Remove the woolly floral parts to lift germination.                 | This guide |
| Division | Not used    | Annual or short-lived.                                              | –          |

Seed collection & storage

| Item                  | Details                                                                     |
|-----------------------|-----------------------------------------------------------------------------|
| Seed collection       | Collect spikes when the flowers are papery and very dry and rub off easily. |
| Extraction & cleaning | Rub spikes over a sieve to separate the individual flowers.                 |
| Seed quality          | Expect many empty flowers – check a few.                                    |
| Storage life          | Not well documented. Store dry and cool.                                    |
| Typical germination   | Not recorded. Untreated <i>Ptilotus</i> seed usually germinates poorly.     |

Sowing & growing on

| Item                  | Details                                                                     |
|-----------------------|-----------------------------------------------------------------------------|
| Best sowing time      | Mar – May. Feb and Jun – Sep are workable.                                  |
| Seed mix              | Standard sterile seed mix, extra free-draining.                             |
| Pre-treatment         | Remove the woolly floral parts to free the seed; water in with smoke water. |
| Sowing depth & method | Cover about 1–2 mm, or sow direct into tubes.                               |
| After sowing          | Keep just moist; good airflow.                                              |
| Germination time      | About 6–10 days for other <i>Ptilotus</i> ; record your own.                |
| Pricking out          | Early and carefully, or sow direct into tubes.                              |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem               | Likely cause and fix            |
|-----------------------|---------------------------------|
| Almost no germination | Wool left on, or empty flowers. |
| Seedlings collapse    | Too wet. Water less.            |
| Weak, floppy plants   | Too much shade. Full sun.       |

### Notes

|             |                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|
| Other notes | Treat as a trial and record results so this page can be updated.                                                                |
| Confidence  | <b>Low.</b> No species data found. Method follows other <i>Ptilotus</i> and research with <i>P. exaltatus</i> – trial protocol. |
| Sources     | See References: 4, 5, 28, 29                                                                                                    |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Rhagodia baccata

Berry Saltbush

Family  
Amaranthaceae  
(chenopods)

| DIFFICULTY | PRE-TREATMENT                        | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|------------|--------------------------------------|-------------|--------------|---------------------|
| Moderate   | Ferment berries;<br>smoke-water soak | Mar – May   | 2–3 mm       | 11%                 |

|                                    | Jan | Feb | Mar  | Apr  | May  | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|------------------------------------|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| Sowing                             | –   | –   | Best | Best | Best | OK  | OK  | OK  | OK  | –   | –   | –   |
| Best window                        |     |     |      |      |      |     |     |     |     |     |     |     |
| Workable – slower or less reliable |     |     |      |      |      |     |     |     |     |     |     |     |
| Avoid / low success                |     |     |      |      |      |     |     |     |     |     |     |     |

## Plant snapshot

|            |                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| Habit      | Spreading shrub, 0.3–2 m, with grey-green leaves.                                                                             |
| Flowering  | Variable – Feb, Apr – May or Oct – Dec, followed by red berries.                                                              |
| Habitat    | Sand dunes and coastal rock.                                                                                                  |
| Local note | Recorded in Joondalup and Wanneroo. FloraBase uses <i>Rhagodia baccata</i> ; national lists use <i>Chenopodium baccatum</i> . |

## Propagation options

| Method   | Suitability | Notes                                                | Details in       |
|----------|-------------|------------------------------------------------------|------------------|
| Cuttings | Good        | Chenopod shrubs generally strike well (provisional). | → Cuttings Guide |
| Seed     | Main method | Ferment berries, then smoke water.                   | This guide       |
| Division | Not used    | Woody shrub.                                         | –                |

## Seed collection & storage

| Item                  | Details                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Pick fully ripe, deep red berries.                                                                                                          |
| Extraction & cleaning | <b>Ferment:</b> soak the berries in water for 4–5 days, wash off the pulp through a sieve and dry the seed.                                 |
| Seed quality          | Small, dark seeds; discard any that float after washing.                                                                                    |
| Storage life          | Not well documented. Store dry and cool.                                                                                                    |
| Typical germination   | About <b>11%</b> (untreated, Alcoa nursery average, 7 tests). About <b>75 seedlings per gram</b> – roughly 1.3 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Mar – May.</b> Jun – Sep workable. Avoid Oct – Feb.                                                                    |
| Seed mix              | Standard sterile native seed mix.                                                                                         |
| Pre-treatment         | Ferment and clean (above), then <b>soak in smoke water (1:9) for 24 hours</b> (Geraldton method for <i>R. preissii</i> ). |
| Sowing depth & method | Cover about <b>2–3 mm</b> . Sow thinly in trays.                                                                          |
| After sowing          | Water in; keep moist, not wet.                                                                                            |
| Germination time      | About <b>7–14 days</b> (Geraldton, <i>R. preissii</i> ).                                                                  |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | When the second pair of leaves appears.                                                                     |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem            | Likely cause and fix                                            |
|--------------------|-----------------------------------------------------------------|
| Poor germination   | Pulp left on, or unripe berries. Ferment fully; use ripe fruit. |
| Mould on trays     | Pulp residue. Wash seed well.                                   |
| Seedlings collapse | Damping off. Sow thinly; good airflow.                          |

### Notes

|             |                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------|
| Other notes | Tip cuttings are worth trialling (see the Cuttings Guide).                                                          |
| Confidence  | <b>Medium.</b> Nursery test data (7 tests); treatment follows the WA community-nursery method for <i>Rhagodia</i> . |
| Sources     | See References: 3, 4, 5, 37                                                                                         |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Spinifex hirsutus

Hairy Spinifex

Family  
Poaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT                               |     |      |      | BEST SOWING      |     |     | SOWING DEPTH              |     | TYPICAL GERMINATION |     |     |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------|-----|------|------|------------------|-----|-----|---------------------------|-----|---------------------|-----|-----|
| <b>Moderate *</b>                                                                                   | <b>Cut off spines/awns;<br/>smoke water</b> |     |      |      | <b>Mar – May</b> |     |     | <b>10 mm (needs dark)</b> |     | <b>Not recorded</b> |     |     |
|                                                                                                     | Jan                                         | Feb | Mar  | Apr  | May              | Jun | Jul | Aug                       | Sep | Oct                 | Nov | Dec |
| <b>Sowing</b>                                                                                       | OK                                          | OK  | Best | Best | Best             | –   | –   | –                         | OK  | OK                  | OK  | OK  |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                                             |     |      |      |                  |     |     |                           |     |                     |     |     |

## Plant snapshot

|                   |                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------|
| <b>Habit</b>      | Rhizomatous, tussock-forming grass, 0.3–0.9 m, with long runners and silvery, hairy leaves. |
| <b>Flowering</b>  | Oct – Jan. Male and female flowers on separate plants; female heads are large spiky balls.  |
| <b>Habitat</b>    | White sand on coastal dunes.                                                                |
| <b>Local note</b> | Recorded in the City of Joondalup.                                                          |

## Propagation options

| Method   | Suitability | Notes                                                              | Details in        |
|----------|-------------|--------------------------------------------------------------------|-------------------|
| Cuttings | Good        | Runner (stolon) sections with 1–2 nodes.                           | → Cuttings Guide  |
| Seed     | Main method | The usual nursery method; cover seed (needs dark).                 | <b>This guide</b> |
| Division | Possible    | Tussock divisions work but survived less well than rooted runners. | → Division Guide  |

## Seed collection & storage

| Item                             | Details                                                                                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Seed collection</b>           | Collect the spiky female heads in summer as they ripen and start to break away. <b>Collect from female plants growing near male plants</b> – isolated females set little seed. |
| <b>Extraction &amp; cleaning</b> | Pull the seed-bearing spikelets from the head and <b>cut off the long spines (awns)</b> .                                                                                      |
| <b>Seed quality</b>              | Many spikelets are empty – squeeze-test a few.                                                                                                                                 |
| <b>Storage life</b>              | Not well documented. Store dry and cool.                                                                                                                                       |
| <b>Typical germination</b>       | Not recorded for this species. In <i>S. sericeus</i> trials, seed germinated far better in darkness (56%) than in light (14%).                                                 |

## Sowing & growing on

| Item                             | Details                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Best sowing time</b>          | <b>Mar – May</b> (provisional). Sep – Feb workable. Avoid Jun – Aug.                            |
| <b>Seed mix</b>                  | Coarse, sandy, free-draining mix.                                                               |
| <b>Pre-treatment</b>             | Cut off the awns; water in with smoke water after sowing (Geraldton method).                    |
| <b>Sowing depth &amp; method</b> | <b>Cover well (about 10 mm)</b> – seed germinates best in the dark. Sow en masse in deep trays. |
| <b>After sowing</b>              | Water in with smoke water; keep moist, not wet.                                                 |
| <b>Germination time</b>          | About <b>10 days</b> (Geraldton nursery).                                                       |
| <b>Pricking out</b>              | When seedlings have 2–3 leaves, into deep tubes.                                                |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Slow – est. 6–9 months from sowing (record your own). Plant out in late autumn or winter.                   |

### Common problems

| Problem            | Likely cause and fix                                                             |
|--------------------|----------------------------------------------------------------------------------|
| Little germination | Seed from isolated females, or sown too shallow. Collect near males; cover seed. |
| Seedlings rot      | Mix too wet. Use coarse sand and water less.                                     |
| Slow growth        | Normal for spinifex. Full sun, deep tubes, light feeding.                        |

### Notes

|             |                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
| Other notes | Rooted runner sections (Cuttings Guide) and tussock division (Division Guide) are alternatives.                         |
| Confidence  | <b>Medium.</b> WA community-nursery method for spinifex plus trial results for the closely related <i>S. sericeus</i> . |
| Sources     | See References: 4, 5, 36                                                                                                |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Spinifex longifolius

Beach Spinifex

Family  
Poaceae

| DIFFICULTY                                                                                    | PRE-TREATMENT                               |                                                                                                                      |      |      | BEST SOWING      |     |     | SOWING DEPTH                                                                                            |     | TYPICAL GERMINATION |     |     |
|-----------------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------|------|------------------|-----|-----|---------------------------------------------------------------------------------------------------------|-----|---------------------|-----|-----|
| <b>Moderate *</b>                                                                             | <b>Cut off spines/awns;<br/>smoke water</b> |                                                                                                                      |      |      | <b>Mar – May</b> |     |     | <b>10 mm (needs dark)</b>                                                                               |     | <b>Not recorded</b> |     |     |
|                                                                                               | Jan                                         | Feb                                                                                                                  | Mar  | Apr  | May              | Jun | Jul | Aug                                                                                                     | Sep | Oct                 | Nov | Dec |
| <b>Sowing</b>                                                                                 | OK                                          | OK                                                                                                                   | Best | Best | Best             | –   | –   | –                                                                                                       | OK  | OK                  | OK  | OK  |
|  Best window |                                             |  Workable – slower or less reliable |      |      |                  |     |     |  Avoid / low success |     |                     |     |     |

## Plant snapshot

|                   |                                                                                          |
|-------------------|------------------------------------------------------------------------------------------|
| <b>Habit</b>      | Rhizomatous, tussock-forming grass, 0.3–1 m high and up to 2 m wide, with long runners.  |
| <b>Flowering</b>  | Apr – Dec or Jan. Male and female plants are separate; female heads up to 250 mm across. |
| <b>Habitat</b>    | White sand on coastal dunes; a key foredune stabiliser.                                  |
| <b>Local note</b> | Recorded in the Wanneroo LGA.                                                            |

## Propagation options

| Method   | Suitability | Notes                                                              | Details in        |
|----------|-------------|--------------------------------------------------------------------|-------------------|
| Cuttings | Good        | Runner (stolon) sections with 1–2 nodes.                           | → Cuttings Guide  |
| Seed     | Main method | The usual nursery method; smoke water, about 10 days.              | <b>This guide</b> |
| Division | Possible    | Tussock divisions work but survived less well than rooted runners. | → Division Guide  |

## Seed collection & storage

| Item                             | Details                                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Seed collection</b>           | Collect ripe female heads as they begin to break away. <b>Choose female plants growing close to males.</b>                     |
| <b>Extraction &amp; cleaning</b> | Separate the seed from the head and <b>cut off the awns.</b>                                                                   |
| <b>Seed quality</b>              | Many spikelets are empty – squeeze-test a few.                                                                                 |
| <b>Storage life</b>              | Not well documented. Store dry and cool.                                                                                       |
| <b>Typical germination</b>       | Not recorded for this species. In <i>S. sericeus</i> trials, seed germinated far better in darkness (56%) than in light (14%). |

## Sowing & growing on

| Item                             | Details                                                                         |
|----------------------------------|---------------------------------------------------------------------------------|
| <b>Best sowing time</b>          | <b>Mar – May</b> (provisional). Sep – Feb workable. Avoid Jun – Aug.            |
| <b>Seed mix</b>                  | Coarse, sandy, free-draining mix.                                               |
| <b>Pre-treatment</b>             | Cut off the awns and <b>sow en masse with smoke water</b> (Geraldton method).   |
| <b>Sowing depth &amp; method</b> | <b>Cover well (about 10 mm)</b> – seed germinates best in the dark. Deep trays. |
| <b>After sowing</b>              | Keep moist, not wet.                                                            |
| <b>Germination time</b>          | About <b>10 days</b> (Geraldton nursery).                                       |
| <b>Pricking out</b>              | When seedlings have 2–3 leaves, into deep tubes.                                |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Slow – est. 6–9 months from sowing (record your own). Plant out in late autumn or winter.                   |

### Common problems

| Problem            | Likely cause and fix                                                             |
|--------------------|----------------------------------------------------------------------------------|
| Little germination | Seed from isolated females, or sown too shallow. Collect near males; cover seed. |
| Seedlings rot      | Mix too wet. Use coarse sand and water less.                                     |
| Slow growth        | Normal for spinifex. Full sun, deep tubes.                                       |

### Notes

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| Other notes | Rooted runner sections (Cuttings Guide) and tussock division (Division Guide) are alternatives.                             |
| Confidence  | <b>Medium.</b> WA community-nursery method for this species plus trial results for the closely related <i>S. sericeus</i> . |
| Sources     | See References: 4, 5, 36, 37                                                                                                |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Spyridium globulosum

Basket Bush

Family  
Rhamnaceae

| DIFFICULTY                                                                                          | PRE-TREATMENT           |     |     | BEST SOWING |      |      | SOWING DEPTH        |     |     | TYPICAL GERMINATION |     |     |
|-----------------------------------------------------------------------------------------------------|-------------------------|-----|-----|-------------|------|------|---------------------|-----|-----|---------------------|-----|-----|
| Moderate                                                                                            | Hot water (or dry heat) |     |     | Apr – Jun   |      |      | 2–3 mm (needs dark) |     |     | 41%                 |     |     |
|                                                                                                     | Jan                     | Feb | Mar | Apr         | May  | Jun  | Jul                 | Aug | Sep | Oct                 | Nov | Dec |
| Sowing                                                                                              | –                       | –   | OK  | Best        | Best | Best | OK                  | OK  | –   | –                   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                         |     |     |             |      |      |                     |     |     |                     |     |     |

## Plant snapshot

|            |                                      |
|------------|--------------------------------------|
| Habit      | Erect shrub, usually 0.6–5 m.        |
| Flowering  | Jun – Nov (white, in woolly heads).  |
| Habitat    | Sand on coastal dunes and limestone. |
| Local note | Recorded in Joondalup and Wanneroo.  |

## Propagation options

| Method   | Suitability | Notes                                          | Details in                 |
|----------|-------------|------------------------------------------------|----------------------------|
| Cuttings | Not used    | Not documented.                                | –                          |
| Seed     | Main method | Hot water; cool, dark conditions to germinate. | <a href="#">This guide</a> |
| Division | Not used    | –                                              | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect fruit as it ripens in late spring–summer; bag branches to catch shed seed.                                                                                                   |
| Extraction & cleaning | Rub and sieve the small seed from the woolly flower heads.                                                                                                                           |
| Seed quality          | Water-impervious seed coat (physical dormancy) – seed won't take up water until treated.                                                                                             |
| Storage life          | Not well documented; hard-coated seed is usually long-lived when dry.                                                                                                                |
| Typical germination   | About <b>41%</b> after heat treatment (Alcoa, only 2 tests). About <b>205 seedlings per gram</b> . Research recorded high germination after hot water at cool (7/18°C) temperatures. |

## Sowing & growing on

| Item                  | Details                                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Apr – Jun</b> – germinates best at winter-like temperatures (7/18°C); warmth suppresses it. Mar, Jul – Aug workable. Avoid Sep – Feb.                   |
| Seed mix              | Standard sterile native seed mix.                                                                                                                          |
| Pre-treatment         | <b>Hot water</b> (about 90–95°C) breaks the seed-coat dormancy – keep it short, as over-long treatment can kill seed. Alcoa used a dry-heat pulse instead. |
| Sowing depth & method | Cover about <b>2–3 mm</b> – germination is suppressed by light, so keep the seed covered and trays shaded.                                                 |
| After sowing          | Water in; keep moist in a cool, shaded spot.                                                                                                               |
| Germination time      | Not well documented – allow several weeks and record results.                                                                                              |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Pricking out            | When the second pair of leaves appears.                                                                     |
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 4–6 months from sowing (record your own).                                                              |

### Common problems

| Problem                  | Likely cause and fix                                                                              |
|--------------------------|---------------------------------------------------------------------------------------------------|
| No germination           | Seed coat not broken, or trays too warm or too light. Re-treat; cover seed; sow in cooler months. |
| Seed killed by treatment | Hot water too long. Shorten the soak.                                                             |
| Damping off              | Wet, crowded trays. Thin sowing, airflow.                                                         |

### Notes

|             |                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------|
| Other notes | One of the few coastal shrubs here that needs cool temperatures <i>and</i> darkness to germinate. |
| Confidence  | <b>High.</b> Species-specific dormancy research and nursery data available.                       |
| Sources     | See References: 3, 4, 22, 23                                                                      |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
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|           |                       |           |          |        |         |       |

# Threlkeldia diffusa

Coast Bonefruit

Family  
Amaranthaceae  
(chenopods)

| DIFFICULTY    | PRE-TREATMENT         | BEST SOWING | SOWING DEPTH | TYPICAL GERMINATION |
|---------------|-----------------------|-------------|--------------|---------------------|
| Easy-Moderate | 24 h smoke-water soak | Apr – Jun   | 1–2 mm       | Not recorded        |

|                                                                                                     | Jan | Feb | Mar | Apr  | May  | Jun  | Jul | Aug | Sep | Oct | Nov | Dec |
|-----------------------------------------------------------------------------------------------------|-----|-----|-----|------|------|------|-----|-----|-----|-----|-----|-----|
| Sowing                                                                                              | –   | –   | OK  | Best | Best | Best | OK  | OK  | –   | –   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |     |     |     |      |      |      |     |     |     |     |     |     |

### Plant snapshot

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| Habit      | Much-branched, prostrate to erect perennial, 0.1–0.4 m, with semi-succulent leaves. |
| Flowering  | Oct – Nov (tiny, green), followed by small red to black fruits.                     |
| Habitat    | White or grey sand over limestone, coastal areas and saline flats.                  |
| Local note | Recorded in the City of Joondalup.                                                  |

### Propagation options

| Method   | Suitability | Notes                                         | Details in       |
|----------|-------------|-----------------------------------------------|------------------|
| Cuttings | Trial       | No published cutting data; trial protocol.    | → Cuttings Guide |
| Seed     | Main method | Smoke-water soak; 10–15 days in cool weather. | This guide       |
| Division | Not used    | Not documented.                               | –                |

### Seed collection & storage

| Item                  | Details                                                         |
|-----------------------|-----------------------------------------------------------------|
| Seed collection       | Collect the small fruits in summer once they turn red to black. |
| Extraction & cleaning | Dry the fruits; sow them whole.                                 |
| Seed quality          | Cut-test a few fruits to check they are filled.                 |
| Storage life          | Not well documented. Store dry and cool.                        |
| Typical germination   | Not recorded.                                                   |

### Sowing & growing on

| Item                  | Details                                                                             |
|-----------------------|-------------------------------------------------------------------------------------|
| Best sowing time      | Apr – Jun – germinates in cooler weather. Mar, Jul – Aug workable. Avoid Sep – Feb. |
| Seed mix              | Standard sterile native seed mix, extra free-draining.                              |
| Pre-treatment         | Soak in smoke water (1:9) for 24 hours (Geraldton method).                          |
| Sowing depth & method | Cover about 1–2 mm.                                                                 |
| After sowing          | Water in; keep just moist – semi-succulent seedlings rot if wet.                    |
| Germination time      | About 10–15 days (Geraldton nursery).                                               |
| Pricking out          | When the second pair of leaves appears.                                             |

| Item                    | Details                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| Shade house & hardening | About <b>2 weeks</b> in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. |
| Time to planting size   | Est. 3–4 months from sowing (record your own).                                                              |

### Common problems

| Problem         | Likely cause and fix                       |
|-----------------|--------------------------------------------|
| No germination  | Too warm. Sow in the cooler months.        |
| Seedlings rot   | Too wet. Water less, improve drainage.     |
| Leggy seedlings | Not enough light. Move to full sun sooner. |

### Notes

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| Other notes | Softwood cuttings are a trial alternative (see the Cuttings Guide).         |
| Confidence  | <b>Medium.</b> WA community-nursery method available; no nursery test data. |
| Sources     | See References: 4, 5                                                        |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# Waitzia suaveolens

Fragrant Waitzia

Family  
Asteraceae

| DIFFICULTY                                                                                          | PRE-TREATMENT          |     |     | BEST SOWING |      |     | SOWING DEPTH          |     |     | TYPICAL GERMINATION |     |     |
|-----------------------------------------------------------------------------------------------------|------------------------|-----|-----|-------------|------|-----|-----------------------|-----|-----|---------------------|-----|-----|
| Easy-Moderate                                                                                       | Smoke water (optional) |     |     | Apr – May   |      |     | Surface – light cover |     |     | 21%                 |     |     |
|                                                                                                     | Jan                    | Feb | Mar | Apr         | May  | Jun | Jul                   | Aug | Sep | Oct                 | Nov | Dec |
| Sowing                                                                                              | –                      | –   | OK  | Best        | Best | OK  | –                     | –   | –   | –                   | –   | –   |
| <div>Best window</div> <div>Workable – slower or less reliable</div> <div>Avoid / low success</div> |                        |     |     |             |      |     |                       |     |     |                     |     |     |

## Plant snapshot

|            |                                                                                          |
|------------|------------------------------------------------------------------------------------------|
| Habit      | Annual herb to 0.6 m.                                                                    |
| Flowering  | Sep – Dec or Jan (white, pink, purple or yellow papery heads).                           |
| Habitat    | Widespread across the south-west, including the Swan Coastal Plain.                      |
| Local note | Recorded in the Wanneroo LGA. An annual – it flowers the spring after sowing, then dies. |

## Propagation options

| Method   | Suitability | Notes                                                  | Details in                 |
|----------|-------------|--------------------------------------------------------|----------------------------|
| Cuttings | Not used    | Annual.                                                | –                          |
| Seed     | Main method | Annual everlasting – sow after the first autumn rains. | <a href="#">This guide</a> |
| Division | Not used    | Annual.                                                | –                          |

## Seed collection & storage

| Item                  | Details                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Seed collection       | Collect heads in early summer once they dry and the fluffy seed starts to lift.                                                              |
| Extraction & cleaning | Rub heads gently over a sieve. Seed is very fine.                                                                                            |
| Seed quality          | Fine seed – a little goes a long way.                                                                                                        |
| Storage life          | Store dry and cool and sow the following autumn.                                                                                             |
| Typical germination   | About <b>21%</b> (untreated, Alcoa nursery average, 2 tests). About <b>967 seedlings per gram</b> – roughly 0.1 g of seed per 100 seedlings. |

## Sowing & growing on

| Item                  | Details                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Best sowing time      | <b>Apr – May</b> , after the first autumn rains. Mar and Jun are workable.                                               |
| Seed mix              | Standard sterile native seed mix.                                                                                        |
| Pre-treatment         | None required. Smoke water in the first watering may help.                                                               |
| Sowing depth & method | Mix the fine seed with dry sand and surface-sow; cover very lightly or not at all. A pinch per tube avoids pricking out. |
| After sowing          | Water in with a fine mist; keep moist.                                                                                   |
| Germination time      | About <b>2–4 weeks</b> .                                                                                                 |
| Pricking out          | If tray-sown, prick out small, at the second-leaf stage. Thin direct-sown cells to 1–3 seedlings.                        |

| Item                    | Details                                                                            |
|-------------------------|------------------------------------------------------------------------------------|
| Shade house & hardening | Short – a week or so in the shade house, then full sun.                            |
| Time to planting size   | Plant out small, about <b>6–10 weeks</b> after sowing, so plants flower in spring. |

### Common problems

| Problem                   | Likely cause and fix                  |
|---------------------------|---------------------------------------|
| Seed washed into clumps   | Bottom water or use a very fine mist. |
| Leggy seedlings           | Too much shade. Full sun early.       |
| Plants flower in the tube | Held too long. Plant out while small. |

### Notes

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| Other notes | Often direct-seeded into revegetation sites after the first autumn rains.                                    |
| Confidence  | <b>Medium.</b> Nursery germination data (2 tests); sowing follows standard practice for annual everlastings. |
| Sources     | See References: 3, 4, 30                                                                                     |

### Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

| Date sown | Seed lot / provenance | Treatment | Qty sown | No. up | Germ. % | Notes |
|-----------|-----------------------|-----------|----------|--------|---------|-------|
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |
|           |                       |           |          |        |         |       |

# References

Numbers match the source list on each species page. Web addresses shortened; accessed September 2026.

| #  | Source                                                                                                                                                                                                                                                                                                            | Address                                                                                                                                                                       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Australian Native Plants Society (ANPSA) – Plant Propagation: seed mix, pre-treatments, pricking out, growing on                                                                                                                                                                                                  | <a href="https://anpsa.org.au/plant-propagation/">anpsa.org.au/plant-propagation/</a>                                                                                         |
| 2  | Cochrane A. (2007) Seed Notes 7 – Allocasuarina. Wildflower Society of WA / DEC Threatened Flora Seed Centre                                                                                                                                                                                                      | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01994.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01994.pdf</a>                                                     |
| 3  | Cromer E.L. (2007) Seed Germination and Research Records from Alcoa's Marrinup Nursery. Alcoa World Alumina Australia                                                                                                                                                                                             | <a href="https://library.dbca.wa.gov.au/static/FullTextFiles/924754.pdf">library.dbca.wa.gov.au/static/FullTextFiles/924754.pdf</a>                                           |
| 4  | FloraBase, WA Herbarium (DBCA) – species profiles 1411, 1728, 1732, 1733, 1800, 1834, 2299, 2301, 2463, 2718, 2742, 2798, 3710, 3824, 3845, 3957, 3966, 4044, 4828, 5426, 5659, 5708, 5920, 8127, 8184, 8282, 11341, 11636, 16449, 17104, 17240, 17657, 18119, 18560, 18598 and those cited in the Cuttings Guide | <a href="https://florabase.dbca.wa.gov.au">florabase.dbca.wa.gov.au</a>                                                                                                       |
| 5  | City of Greater Geraldton Community Nursery – Native Plant Propagation Guide (2019): seed treatments, smoke water, damping off                                                                                                                                                                                    | <a href="https://cgg.wa.gov.au">cgg.wa.gov.au</a>                                                                                                                             |
| 6  | Seed Notes 3 – Seed quality and germination. Wildflower Society of WA / DEC (2007)                                                                                                                                                                                                                                | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01990.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01990.pdf</a>                                                     |
| 7  | Seed Science Research – Seed ageing of four Western Australian species in relation to storage environment (incl. <i>A. manglesii</i> )                                                                                                                                                                            | <a href="https://cambridge.org">cambridge.org</a> (Seed Science Research)                                                                                                     |
| 8  | AoB PLANTS (2018) – Seed dormancy regulates germination response to smoke and temperature ( <i>Anigozanthos</i> )                                                                                                                                                                                                 | <a href="https://academic.oup.com/aobpla/article/10/4/ply042/5048955">academic.oup.com/aobpla/article/10/4/ply042/5048955</a>                                                 |
| 9  | Wikipedia – <i>Anigozanthos manglesii</i> (cultivation, ink disease)                                                                                                                                                                                                                                              | <a href="https://en.wikipedia.org/wiki/Anigozanthos_manglesii">en.wikipedia.org/wiki/Anigozanthos_manglesii</a>                                                               |
| 10 | ANPSA – The Propagation of <i>Banksia</i> (extraction, mix, damping off, low-P feeding, hardening)                                                                                                                                                                                                                | <a href="https://anpsa.org.au/the-propagation-of-banksia/">anpsa.org.au/the-propagation-of-banksia/</a>                                                                       |
| 11 | Seed Notes 8 – <i>Banksia</i> . Wildflower Society of WA / DEC (2007)                                                                                                                                                                                                                                             | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01995.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01995.pdf</a>                                                     |
| 12 | Wikipedia – <i>Banksia attenuata</i> and <i>Banksia menziesii</i> (germination times, damping off, serotiny)                                                                                                                                                                                                      | <a href="https://en.wikipedia.org">en.wikipedia.org</a>                                                                                                                       |
| 13 | Crawford A.D. – Seed storage and longevity of Australian plant species (PhD thesis; <i>Banksia</i> cone age and viability)                                                                                                                                                                                        | <a href="https://library.dbca.wa.gov.au/static/FullTextFiles/924221.pdf">library.dbca.wa.gov.au/static/FullTextFiles/924221.pdf</a>                                           |
| 14 | Seed Notes 15 – Hard seed: <i>Chorizema</i> , <i>Daviesia</i> , <i>Gastrolobium</i> and <i>Gompholobium</i> . Wildflower Society of WA / DEC                                                                                                                                                                      | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM02002.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM02002.pdf</a>                                                     |
| 15 | Seed Notes 9 – <i>Acacia</i> (hard-seed longevity, float test). Wildflower Society of WA / DEC                                                                                                                                                                                                                    | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01996.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01996.pdf</a>                                                     |
| 16 | Auld T.D. (1996) Ecology of the Fabaceae: fire, ants and the soil seedbank. <i>Cunninghamia</i> 4(4)                                                                                                                                                                                                              | <a href="https://botanicgardens.org.au">botanicgardens.org.au</a>                                                                                                             |
| 17 | Seed Notes 2 – Seed collection and storage. Wildflower Society of WA / DEC (2007)                                                                                                                                                                                                                                 | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01989.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01989.pdf</a>                                                     |
| 18 | ANPSA – <i>Melaleuca</i> genus page (seed from old capsules, no pre-treatment, bog method, 14–30 days)                                                                                                                                                                                                            | <a href="https://anpsa.org.au/genera/melaleuca/">anpsa.org.au/genera/melaleuca/</a>                                                                                           |
| 19 | Seed Notes 6 – <i>Eucalyptus</i> . Wildflower Society of WA / DEC (2007)                                                                                                                                                                                                                                          | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01993.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01993.pdf</a>                                                     |
| 20 | Seed Notes 13 – <i>Dryandra</i> (now <i>Banksia</i> ). Wildflower Society of WA / DEC (2007)                                                                                                                                                                                                                      | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM02000.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM02000.pdf</a>                                                     |
| 21 | Seed Notes 18 – <i>Isopogon</i> and <i>Petrophile</i> . Wildflower Society of WA / DEC (2007)                                                                                                                                                                                                                     | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM02005.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM02005.pdf</a>                                                     |
| 22 | Seed Science Research – Physical dormancy in seeds of six genera of Australian <i>Rhamnaceae</i> (incl. <i>Spyridium globulosum</i> )                                                                                                                                                                             | <a href="https://cambridge.org">cambridge.org</a> (Seed Science Research)                                                                                                     |
| 23 | Florabank Guidelines Module 11 – Seed germination and dormancy. Australian Network for Plant Conservation                                                                                                                                                                                                         | <a href="https://anpc.asn.au/wp-content/uploads/2025/08/11-Seed-Germination-and-Dormancy.pdf">anpc.asn.au/wp-content/uploads/2025/08/11-Seed-Germination-and-Dormancy.pdf</a> |

| #  | Source                                                                                                                    | Address                                                                                                                                                                                 |
|----|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | ANPSA – Kennedia prostrata profile                                                                                        | <a href="https://anpsa.org.au/plant_profiles/kennedia-prostrata/">anpsa.org.au/plant_profiles/kennedia-prostrata/</a>                                                                   |
| 25 | Wikipedia – Banksia sessilis (seed release, fire response, cuttings)                                                      | <a href="https://en.wikipedia.org/wiki/Banksia_sessilis">en.wikipedia.org/wiki/Banksia_sessilis</a>                                                                                     |
| 26 | Conservation Physiology (2023) – Are current seed storage approaches suitable for Macrozamia fraseri?                     | <a href="https://academic.oup.com/conphys/article/11/1/coad096/7471550">academic.oup.com/conphys/article/11/1/coad096/7471550</a>                                                       |
| 27 | Wikipedia – Macrozamia fraseri (description, distribution, toxicity)                                                      | <a href="https://en.wikipedia.org/wiki/Macrozamia_fraseri">en.wikipedia.org/wiki/Macrozamia_fraseri</a>                                                                                 |
| 28 | Scientia Horticulturae (1989) – Cultivation of the pink mulla mulla, Ptilotus exaltatus. 1. Seed germination and dormancy | <a href="https://sciencedirect.com/science/article/abs/pii/0304423889901192">sciencedirect.com/science/article/abs/pii/0304423889901192</a>                                             |
| 29 | ANPSA – Ptilotus exaltatus profile (seed slow and unreliable; cuttings successful)                                        | <a href="https://anpsa.org.au/plant_profiles/ptilotus-exaltatus/">anpsa.org.au/plant_profiles/ptilotus-exaltatus/</a>                                                                   |
| 30 | ANPSA – Australian Daisies (annuals from seed, light, smoke and GA; perennial cuttings)                                   | <a href="https://anpsa.org.au/genera/daisies/">anpsa.org.au/genera/daisies/</a>                                                                                                         |
| 31 | Australian National Botanic Gardens – Leucophyta brownii (no pre-treatment; 10–30 days)                                   | <a href="https://anbg.gov.au/gnp/interns-2014/leucophyta-brownii.html">anbg.gov.au/gnp/interns-2014/leucophyta-brownii.html</a>                                                         |
| 32 | Australian National Botanic Gardens – Carpobrotus glaucescens (grown from seed or cuttings)                               | <a href="https://anbg.gov.au/gnp/interns-2005/carpobrotus-glaucescens.html">anbg.gov.au/gnp/interns-2005/carpobrotus-glaucescens.html</a>                                               |
| 33 | Australian National Botanic Gardens – Atriplex nummularia (wash or soak fruits; sow autumn, early winter or spring)       | <a href="https://anbg.gov.au/gnp/interns-2007/atriplex-nummularia.html">anbg.gov.au/gnp/interns-2007/atriplex-nummularia.html</a>                                                       |
| 34 | Seed Notes 11 – Grevillea. Wildflower Society of WA / DEC (2007)                                                          | <a href="https://library.dbca.wa.gov.au/FullTextFiles/PAM01998.pdf">library.dbca.wa.gov.au/FullTextFiles/PAM01998.pdf</a>                                                               |
| 35 | ANPSA – Grevillea vestita profile (grows reasonably easily from seed)                                                     | <a href="https://anpsa.org.au/plant_profiles/grevillea-vestita/">anpsa.org.au/plant_profiles/grevillea-vestita/</a>                                                                     |
| 36 | NZ Department of Conservation – Rehabilitation of coastal foredunes (Spinifex sericeus seed in dark vs light; division)   | <a href="https://doc.govt.nz/Documents/science-and-technical/sfc122.pdf">doc.govt.nz/Documents/science-and-technical/sfc122.pdf</a>                                                     |
| 37 | Cottesloe Coastcare Association – plant pages (Rhagodia baccata, Spinifex longifolius)                                    | <a href="https://cottesloecoastcare.org">cottesloecoastcare.org</a>                                                                                                                     |
| 38 | Nurseries Online – Atriplex nummularia (seed soaked, sown 1–2 mm deep)                                                    | <a href="https://nurseriesonline.com.au/plant-index/australian-native-plants/atriplex-nummularia/">nurseriesonline.com.au/plant-index/australian-native-plants/atriplex-nummularia/</a> |

## Revision log

| Version | Date     | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | Sep 2026 | First edition: Allocasuarina fraseriana, A. humilis, Anigozanthos manglesii, Banksia attenuata, B. menziesii, Daviesia nudiflora, Gompholobium tomentosum, Hovea pungens.                                                                                                                                                                                                                                                                                              |
| 1.1     | Sep 2026 | Added 15 species: Austrostipa flavescens, Banksia sessilis, Bossiaea eriocarpa, Calothamnus quadrifidus, Corymbia calophylla, Dianella revoluta, Eucalyptus gomphocephala, E. marginata, Kennedia prostrata, Melaleuca huegelii, M. systema, Olearia axillaris, Petrophile linearis, P. macrostachya, Spyridium globulosum.                                                                                                                                            |
| 1.2     | Sep 2026 | Added 18 species: Allocasuarina lehmanniana, Atriplex isatidea, Carpobrotus virescens, Daviesia divaricata, D. triflora, Grevillea vestita, Leucophyta brownii, Macrozamia fraseri, Myoporum insulare, Podothea gnaphalioides, Ptilotus drummondii, P. manglesii, P. polystachyus, Rhagodia baccata, Spinifex hirsutus, S. longifolius, Threlkeldia diffusa, Waitzia suaveolens. Added a 'Propagation options' table (cuttings, seed, division) to every species page. |
|         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|         |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |